

OPTIMUM

Clinical Trial Report

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Table of contents

1	Introduction	4
1.1	Purpose	4
1.2	Interventions	4
1.3	Trial design	4
1.4	Trial outcomes	5
1.5	Analyses	6
2	Enrolment, Characteristics and Completion	7
2.1	Enrolment	7
2.2	Characteristics	8
2.3	Study completion	10
2.4	Missed visits	12
2.5	Missed food and household questionnaires	13
3	Outcomes	14
3.1	Skin prick tests	14
3.2	Food allergy	22
3.3	Eczema	25
4	Safety	28
4.1	Serious adverse events	28
4.2	Adverse events	29
4.3	Diary Cards	33
5	Appendix	44
5.1	Primary outcome listings	44
5.2	Secondary outcome listings	46
5.3	IgE responses	48

List of Figures

1	Cumulative enrolment.	7
2	Age at skin prick test by protocol stage and occasion type (scheduled or unscheduled). Y-axis scale varies by panel.	15

3	Posterior density for difference in probability of positive skin prick test by size, unadjusted primary model.	20
4	Posterior difference in probability of skin prick test sensitisation for wP compared to aP, adjusted model.	21
5	Posterior density for difference in probability of primary outcome, unadjusted primary model.	24
6	Posterior difference in probability of food allergy for wP compared to aP, adjusted model.	24
7	Posterior density for difference in probability of eczema diagnosis by age at reaction, unadjusted primary model.	26
8	Posterior difference in probability of eczema diagnosis for wP compared to aP, adjusted model.	27
9	Irritability day 0 to 6 by vaccination age and assigned treatment.	34
10	Vomiting day 0 to 6 by vaccination age and assigned treatment.	35
11	Diarrhoea day 0 to 6 by vaccination age and assigned treatment.	36
12	Decreased feeding day 0 to 6 by vaccination age and assigned treatment.	37
13	Drowsiness day 0 to 6 by vaccination age and assigned treatment.	38
14	Restlessness day 0 to 6 by vaccination age and assigned treatment.	39
15	Fever by vaccination age and assigned treatment, stage 1 and 2 combined.	40
16	Injection site erythema by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.	41
17	Injection site swelling by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.	42
18	Injection site induration by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.	42
19	Injection site pain by vaccination age, vaccine, and assigned treatment.	43

List of Tables

1	Baseline characteristics of trial participants.	8
2	Number of early study withdrawals and associated reasons.	10
3	Number of early study withdrawals and associated reasons by study site.	10
4	Listing of infants with last contact occurring at or prior to 15 months of age.	11
5	Number and proportion of infants who attended each visit in stage 1 (based on presence of “visit date”). 6-month + 72 hour was optional visit.	12
6	Number and proportion of infants who attended each visit in stage 2 (based on presence of “visit date”). 19-month visit was for first 150 at PCH only.	12
7	Number and proportion of infants who completed food and household questionnaire for each occasion in stage 1.	13
8	Number and proportion of infants who completed food and household questionnaire for each occasion in stage 2.	13
9	Listing of infants with no SPT records.	14
10	Number and percentage of infants with scheduled and unscheduled skin prick tests.	14
11	Listing of SPTs after 18-months of age.	16
12	Number and percentage of infants with a positive SPT by 18 months of age. Infants with no SPT records counted as missing.	16
13	Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen (for those included in the standard panel). Each cell gives positive/tested (%) [missing result%].	17

14	Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen for additional allergens which were tested. Each cell gives positive/tested (%). Totals restricted to only those infants with at least one additional allergen tested	17
15	Posterior summaries for probability and difference in probability of positive skin prick test by size, unadjusted primary model. Point indicates median.	20
16	Number and percentage of infants who met each primary outcome condition and treatment group.	23
17	Number and percentage of infants who met the primary outcome by primary reason and treatment group.	23
18	Posterior summaries for probability and difference in probability of food allergy, unadjusted primary model.	24
19	Number and percentage of infants with any clinician-diagnosed eczema.	25
20	Number and percentage of infants with new-onset clinician-diagnosed eczema. To be new-onset the first reaction date needed to be after the infants randomisation date. Those with missing were due to eczema diagnosis being present but no SPT records being available.	25
21	Posterior summaries for probability and difference in probability of eczema by age of onset, unadjusted primary model.	26
22	Listing of serious adverse events. Missing values were due to no adverse event record for corresponding SAE log record.	28
23	Number of adverse events by study stage and site.	29
24	Adverse event listing of infants randomised at a site other than PCH.	29
25	Number and proportion of infants with any adverse events and the number of adverse events per infant, at PCH.	29
26	Listing of adverse events designated as "Related".	30
27	Proportion of infants with any event by MedDRA preferred terms, PCH only.	30
28	Number of events by MedDRA preferred terms, PCH only.	31
29	Returned diary cards from stage 1.	33
30	Returned diary cards from stage 2.	33
31	Listing of infants with oral food challenge outcome by 18 months of age and corresponding SPT positive allergens.	44
32	Listing of infants with anaphylaxis outcome by 18 months of age and corresponding SPT positive allergens.	44
33	Listing of infants with IgE-mediated food allergy outcome by 18 months of age and corresponding SPT positive allergens.	45
34	Listing of infants with eczema diagnosis and positive SPT to any allergen.	46
35	Sample IgE concentration summaries (kU/L) by allergen type, visit age, and assigned vaccination schedule.	48
36	Geometric mean ratio (GMR) and geometric mean fold rise ratio (GMFRR) for total IgE.	49
37	Summary of sensitisation model by response type.	49
38	Summary of sensitisation model for tetanus toxoid IgE.	50

1 Introduction

1.1 Purpose

This report summarises the data and results of the analyses for main outcomes of the OPTIMUM trial. The report includes results for:

- baseline demographics
- skin prick tests
- food allergy
- eczema
- safety reporting

For full details of the trial processes, objectives, and outcomes refer to the protocol and statistical analysis plan.

1.2 Interventions

Participants were randomised 1:1 to receive either:

- Combined Diphtheria (D)-Tetanus (T)-acellular Pertussis (aP), Hepatitis B (HepB), Inactivated Poliovirus (IPV) and Haemophilus influenzae type b (Hib) vaccine (DTaPHepB- IPV-Hib) administered at approximately 2 months, together with the standard schedule of vaccines (referred to as 3-aP + aP or aP for short), OR
- Combined Diphtheria (D)-Tetanus (T)-whole cell Pertussis (wP), Hepatitis B (HepB) and Haemophilus influenzae type b (Hib) vaccine (DTwP-HepB- Hib) in place of DTaP-HepB-IPV-Hib at approximately 2 months, together with the standard schedule of vaccines (referred to as 1-wP/2-aP + aP or wP for short).

1.3 Trial design

Originally, a sample size of 3,000 participants was planned. Under the adaptive design, as outlined in the protocol and statistical analysis plan, we estimated the trial had 85% power to detect a reduction in food allergy from 10% to 7% with type I error of 5%. If δ denotes the difference in probability of food allergy under each intervention, then trial success required that δ was very likely to be less than zero, specifically that $\Pr(\delta < 0) > 0.95$.

The maximum sample size was subsequently revised down to 2,000 participants due to feasibility constraints. The trial was stopped early due to logistical issues with 972 participants randomised.

1.4 Trial outcomes

1.4.1 Primary outcome (1)

IgE mediated food allergy with evidence of food sensitisation on skin prick test (SPT) by 12 months old and confirmed (where necessary) by medically supervised oral food challenge(s) (OFC). For the purpose of this trial, if either of the following conditions are met, the study participant will be considered to have reached the primary endpoint:

1. Unequivocal evidence of IgE mediated food allergy, defined as:
 - i. A positive OFC with evidence of sensitisation on SPT to the food of interest *OR*
 - ii. History of clinician-confirmed food anaphylaxis, with symptoms affecting at least two of the following: skin, gastrointestinal tract, respiratory tract, cardiovascular system; *AND* evidence of sensitisation on SPT to the food of interest.
2. Highly probable IgE-mediated food allergy defined as a history of food allergic reaction (consistent with PRACTALL criteria), with evidence of sensitisation on SPT to the food of interest.

We will report on the number and proportion who satisfy this composite definition, and the number and proportion satisfying each of the three sub-components as per a hierarchy in order of 1.i, 1.ii, 2.

1.4.2 Secondary outcomes (2)

- a. A history of parent-reported clinician-diagnosed new onset eczema by **6 months** *AND* a positive SPT to any allergen by approximately 12 months old.
- b. A history of parent-reported clinician-diagnosed new onset eczema by **12 months** *AND* a positive SPT to any allergen by approximately 12 months old.
- c. A SPT **>1mm** greater than the negative control to at least one allergen by approximately 12 months of age
- d. A SPT **≥3mm** greater than the negative control to at least one allergen by approximately 12 months of age

1.4.3 Further Details

The protocol defined a panel of allergens to be tested. In addition to these, further allergens could also be tested at parental request or at discretion of the site PI. When the outcomes say “at least one allergen” [i.e. for eczema for positive SPT] we mean any allergen, including any additional allergens tested beyond the panel defined in the protocol.

For both food allergy and eczema, sensitisation on (positive) SPT was defined to be a result more than 1mm greater than negative control by 18 months of age. For food allergy, the positive SPT needed to match the specific food allergen.

For eczema, the age requirement (by 6 or 12 months of age) was determined by the “age at which allergy first occurred” (outageval). To be considered “new-onset” eczema, this age at which allergy first occurred needed to be after the infants enrolment date. Infants with clinician-diagnosed eczema first occurring prior to enrolment were to be excluded as it’s not possible for them to have “new-onset eczema”.

1.5 Analyses

1.5.1 Primary outcome

The primary analysis was to assume independent Binomial models for occurrence of food allergy under each treatment group with uniform priors on the outcome probabilities.

i Primary model

Let θ_{wP} and θ_{aP} denote the probability of the outcome amongst infants who follow the respective strategy and $\delta = \theta_{wP} - \theta_{aP}$ the difference in probability. If n_{wP} and n_{aP} infants were assigned to each respective strategy, with y_{wP} and y_{aP} events observed, then from our assumed model we estimate that $\theta_j \sim \text{Beta}(1 + y_j, 1 + n_j - y_j)$ represents our uncertainty and δ the difference between these.

A secondary analysis was to assume a logistic regression model for occurrence of food allergy adjusting for baseline factors: site, sex, breastfeeding status, birth order, parental income, mode of delivery, and family history of atopic disease.

1.5.2 Secondary clinical outcomes

Analyses of secondary outcomes are analogous to those for the primary outcome, both the primary and secondary models.

2 Enrolment, Characteristics and Completion

2.1 Enrolment

There were 972 participants randomised, 486 per treatment group. The first participant was randomised on 2018-03-07 and the final participant was randomised on 2023-12-22. A summary of enrolment over calendar time is presented in Figure 1

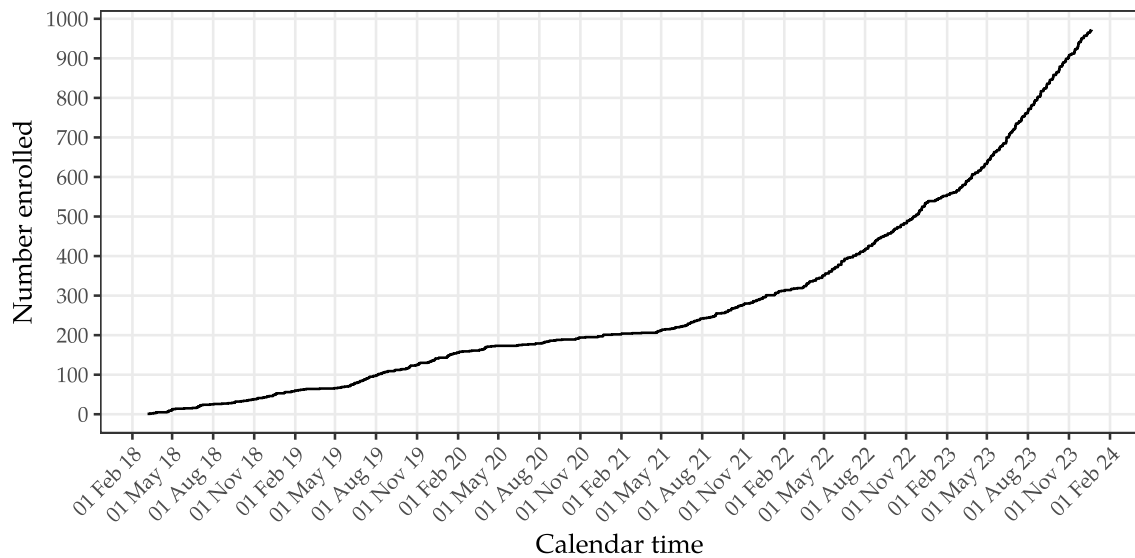


Figure 1: Cumulative enrolment.

2.2 Characteristics

The baseline characteristics of participants by their randomly assigned treatment group are reported in Table 1. Characteristics were similar between the groups.

Table 1: Baseline characteristics of trial participants.

Characteristic	N	aP N = 486 ¹	wP N = 486 ¹	Overall N = 972 ¹
Protocol stage	972			
1		75 (15%)	75 (15%)	150 (15%)
2		411 (85%)	411 (85%)	822 (85%)
Participating site	972			
CHW		30 (6.2%)	29 (6.0%)	59 (6.1%)
MCRI		220 (45%)	219 (45%)	439 (45%)
PCH		206 (42%)	207 (43%)	413 (42%)
SCHN		30 (6.2%)	31 (6.4%)	61 (6.3%)
Gender	972			
Female		221 (45%)	242 (50%)	463 (48%)
Male		265 (55%)	244 (50%)	509 (52%)
Age at enrolment (days)	972	47.0 (44.0, 50.0)	47.0 (44.0, 50.0)	47.0 (44.0, 50.0)
Combined parental income status	971			
< \$18,000		1 (0.2%)	0 (0%)	1 (0.1%)
\$18,000 to \$37,000		1 (0.2%)	4 (0.8%)	5 (0.5%)
\$37,001 to \$87,000		22 (4.5%)	20 (4.1%)	42 (4.3%)
\$87,001 to \$180,000		197 (41%)	222 (46%)	419 (43%)
>\$180,000		264 (54%)	240 (49%)	504 (52%)
(Missing)		1	0	1
First born	972	278 (57%)	273 (56%)	551 (57%)
Child is currently being	972			
Both breastfed and formula-fed		111 (23%)	140 (29%)	251 (26%)
Both breastfed and started on solids		1 (0.2%)	0 (0%)	1 (0.1%)
Exclusively Breastfed		343 (71%)	321 (66%)	664 (68%)
Exclusively formula-fed		31 (6.4%)	25 (5.1%)	56 (5.8%)
Cat ownership	972			
No		373 (77%)	374 (77%)	747 (77%)
Inside		57 (12%)	62 (13%)	119 (12%)
Outside		2 (0.4%)	3 (0.6%)	5 (0.5%)
Both		54 (11%)	47 (9.7%)	101 (10%)
Dog ownership	971			
No		239 (49%)	278 (57%)	517 (53%)
Inside		57 (12%)	48 (9.9%)	105 (11%)
Outside		12 (2.5%)	10 (2.1%)	22 (2.3%)
Both		177 (36%)	150 (31%)	327 (34%)
(Missing)		1	0	1
Maternal gravidity	971			
1		213 (44%)	205 (42%)	418 (43%)
2		144 (30%)	154 (32%)	298 (31%)
3		69 (14%)	70 (14%)	139 (14%)
4		24 (4.9%)	32 (6.6%)	56 (5.8%)
5		16 (3.3%)	5 (1.0%)	21 (2.2%)
6+		19 (3.9%)	20 (4.1%)	39 (4.0%)
(Missing)		1	0	1
Maternal parity	971			
1		284 (59%)	279 (57%)	563 (58%)
2		150 (31%)	166 (34%)	316 (33%)
3		41 (8.5%)	30 (6.2%)	71 (7.3%)
4+		10 (2.1%)	11 (2.3%)	21 (2.2%)
(Missing)		1	0	1

Delivery type	971			
Elective caesarean section		116 (24%)	121 (25%)	237 (24%)
Emergency caesarean section		97 (20%)	108 (22%)	205 (21%)
Forceps/ vacuum assisted delivery		52 (11%)	56 (12%)	108 (11%)
Forceps/vacuum assisted delivery		17 (3.5%)	11 (2.3%)	28 (2.9%)
Vaginal delivery		204 (42%)	189 (39%)	393 (40%)
(Missing)		0	1	1
Intrapartum Antibiotics?	962	166 (35%)	146 (30%)	312 (32%)
(Missing)		6	4	10
Neonatal systemic antibiotics (first 28 days of life)?	971	41 (8.5%)	35 (7.2%)	76 (7.8%)
(Missing)		1	0	1
Gestational age at delivery (weeks)	972	39.00 (38.00, 40.00)	39.00 (38.00, 39.00)	39.00 (38.00, 39.00)
Weight (grams)	970	3,420 (3,140, 3,750)	3,400 (3,100, 3,760)	3,405 (3,120, 3,752)
(Missing)		2	0	2
Length (cm)	965	50.50 (49.00, 52.00)	50.00 (49.00, 52.00)	50.00 (49.00, 52.00)
(Missing)		4	3	7
Head Circumference (cm)	963	35.00 (34.00, 36.00)	35.00 (34.00, 36.00)	35.00 (34.00, 36.00)
(Missing)		5	4	9
Apgar score (1 minute)	963			
1-7		56 (12%)	65 (13%)	121 (13%)
8		51 (11%)	67 (14%)	118 (12%)
9		372 (78%)	351 (73%)	723 (75%)
10		1 (0.2%)	0 (0%)	1 (0.1%)
(Missing)		6	3	9
Apgar score (5 minutes)	963			
1-7		6 (1.3%)	10 (2.1%)	16 (1.7%)
8		14 (2.9%)	20 (4.1%)	34 (3.5%)
9		434 (90%)	429 (89%)	863 (90%)
10		26 (5.4%)	24 (5.0%)	50 (5.2%)
(Missing)		6	3	9
Maternal pertussis vaccination during preceding pregnancy?	971	465 (96%)	459 (95%)	924 (95%)
(Missing)		0	1	1
Maternal pertussis Vaccine administered	909			
Adacel		169 (37%)	161 (35%)	330 (36%)
Boostrix		238 (52%)	240 (53%)	478 (53%)
Unknown		48 (11%)	53 (12%)	101 (11%)
(Missing)		31	32	63
Other maternal pertussis vaccination (last 5 years)	968	240 (50%)	238 (49%)	478 (49%)
(Missing)		2	2	4
Maternal seasonal influenza vaccination during pregnancy?	966	381 (79%)	369 (77%)	750 (78%)
(Missing)		1	5	6
Family history of atopy	972	418 (86%)	415 (85%)	833 (86%)

¹n (%); Median (Q1, Q3)

2.3 Study completion

The study completion status as reported in the database is summarised in Table 2 and by site in Table 3. Most of the *early termination - consent withdrawal* and *early termination - protocol violation/deviation* were due to the first 300 PCH participants who were scheduled for the additional blood collections, a number of which withdrew consent for blood collection at the 18-month and 19-month visits (Table 3). Data was generally still collected for these participants with the exception of these blood collections.

Table 2: Number of early study withdrawals and associated reasons.

Participant status	Reason	aP N = 486	wP N = 486	Overall N = 972
Completed protocol	—	446 (91.8%)	454 (93.4%)	900 (92.6%)
Early Termination	Consent withdrawal	12 (2.5%)	12 (2.5%)	24 (2.5%)
Early Termination	Lost to follow-up	6 (1.2%)	4 (0.8%)	10 (1.0%)
Early Termination	Moved from study area	7 (1.4%)	6 (1.2%)	13 (1.3%)
Early Termination	Protocol violation/deviation	15 (3.1%)	10 (2.1%)	25 (2.6%)

Table 3: Number of early study withdrawals and associated reasons by study site.

Participant status	Reason	aP N = 486	wP N = 486	Overall N = 972
CHW				
Completed protocol	—	30 (100.0%)	29 (100.0%)	59 (100.0%)
MCRI				
Completed protocol	—	216 (98.2%)	216 (98.6%)	432 (98.4%)
Early Termination	Consent withdrawal	2 (0.9%)	— (—)	2 (0.5%)
Early Termination	Lost to follow-up	1 (0.5%)	1 (0.5%)	2 (0.5%)
Early Termination	Moved from study area	1 (0.5%)	2 (0.9%)	3 (0.7%)
PCH				
Completed protocol	—	172 (83.5%)	181 (87.4%)	353 (85.5%)
Early Termination	Consent withdrawal	10 (4.9%)	12 (5.8%)	22 (5.3%)
Early Termination	Lost to follow-up	4 (1.9%)	2 (1.0%)	6 (1.5%)
Early Termination	Moved from study area	5 (2.4%)	2 (1.0%)	7 (1.7%)
Early Termination	Protocol violation/deviation	15 (7.3%)	10 (4.8%)	25 (6.1%)
SCHN				
Completed protocol	—	28 (93.3%)	28 (90.3%)	56 (91.8%)
Early Termination	Lost to follow-up	1 (3.3%)	1 (3.2%)	2 (3.3%)
Early Termination	Moved from study area	1 (3.3%)	2 (6.5%)	3 (4.9%)

There were 17 participants where contact was lost at or before 15 months of age (based on discdat) (Table 4).

Table 4: Listing of infants with last contact occurring at or prior to 15 months of age.

Record ID	Withdrawal reason	Date of Birth	Age at last contact	Attend 12-month visit	Attend 18-month visit
5785-110	Lost to follow-up	2022-05-08	1	FALSE	FALSE
5785-132	Consent withdrawal	2022-06-25	3	FALSE	FALSE
30	Consent withdrawal	2018-08-06	6	FALSE	FALSE
5785-404	Consent withdrawal	2023-09-06	8	FALSE	FALSE
59	Lost to follow-up	2018-12-18	9	FALSE	FALSE
4629-177	Consent withdrawal	2023-01-28	11	FALSE	FALSE
141	Consent withdrawal	2019-10-21	12	FALSE	FALSE
4629-150	Moved from study area	2022-10-25	12	TRUE	FALSE
5785-433	Moved from study area	2023-10-23	12	TRUE	FALSE
8644-18	Moved from study area	2023-04-07	12	FALSE	FALSE
4629-184	Moved from study area	2023-02-15	13	FALSE	FALSE
8644-22	—	2023-05-01	13	FALSE	FALSE
8644-28	Lost to follow-up	2023-05-22	13	FALSE	FALSE
4629-65	Consent withdrawal	2021-04-07	14	FALSE	FALSE
8644-12	Moved from study area	2023-03-28	14	TRUE	FALSE
75	Consent withdrawal	2019-04-18	15	FALSE	FALSE
4629-232	Consent withdrawal	2023-07-31	15	FALSE	FALSE

2.4 Missed visits

Table 5: Number and proportion of infants who attended each visit in stage 1 (based on presence of “visit date”). 6-month + 72 hour was optional visit.

Visit	aP	aP	Overall
	N = 75	N = 75	N = 150
4-month	75 (100%)	75 (100%)	150 (100%)
6-month	75 (100%)	74 (99%)	149 (99%)
6-month + 72 hrs	32 (43%)	25 (33%)	57 (38%)
7-month	73 (97%)	74 (99%)	147 (98%)
12-month	70 (93%)	73 (97%)	143 (95%)
18-month	72 (96%)	71 (95%)	143 (95%)
19-month	70 (93%)	69 (92%)	139 (93%)

Table 6: Number and proportion of infants who attended each visit in stage 2 (based on presence of “visit date”). 19-month visit was for first 150 at PCH only.

Visit	aP	aP	Overall
	N = 411	N = 411	N = 822
12-month	400 (97%)	404 (98%)	804 (98%)
18-month	399 (97%)	398 (97%)	797 (97%)
19-month	58 (14%)	46 (11%)	104 (13%)

2.5 Missed food and household questionnaires

Table 7: Number and proportion of infants who completed food and household questionnaire for each occasion in stage 1.

Visit	aP N = 75	aP N = 75	Overall N = 150
6-week	75 (100%)	75 (100%)	150 (100%)
4-month	75 (100%)	75 (100%)	150 (100%)
6-month	75 (100%)	74 (99%)	149 (99%)
7-month	73 (97%)	74 (99%)	147 (98%)
9-month	74 (99%)	74 (99%)	148 (99%)
12-month	70 (93%)	73 (97%)	143 (95%)
15-month	70 (93%)	73 (97%)	143 (95%)
18-month	71 (95%)	71 (95%)	142 (95%)
19-month	70 (93%)	69 (92%)	139 (93%)

Table 8: Number and proportion of infants who completed food and household questionnaire for each occasion in stage 2.

Visit	aP N = 411	aP N = 411	Overall N = 822
6-week	411 (100%)	411 (100%)	822 (100%)
6-month	405 (99%)	401 (98%)	806 (98%)
9-month	406 (99%)	400 (97%)	806 (98%)
12-month	404 (98%)	404 (98%)	808 (98%)
15-month	394 (96%)	384 (93%)	778 (95%)
18-month	401 (98%)	399 (97%)	800 (97%)

3 Outcomes

All of the trial outcomes depend upon a positive SPT to either a specific allergen or to any allergen. Therefore, this section first reports on the SPT data, and outcomes 2c and 2d, followed by the primary (food allergy) and other secondary (eczema) outcomes.

3.1 Skin prick tests

There were 23 participants with no documented SPTs (Table 9). In many cases this was due to early withdrawal from the study, with the remaining due to non-consent from the parent. All other participants had a record for at least the scheduled SPT, and in a few cases additional unscheduled SPT (Table 10).

Table 9: Listing of infants with no SPT records.

Record ID	rand_site	Reason for ending study participation	Last contact age (months)
30	PCH	Early Termination	6
59	PCH	Early Termination	9
141	PCH	Early Termination	12
4629-65	PCH	Early Termination	14
4629-120	PCH	Early Termination	19
4629-177	PCH	Early Termination	11
4629-184	PCH	Early Termination	13
4629-232	PCH	Early Termination	15
4630-15	CHW	Completed protocol	18
5785-110	MCRI	Early Termination	1
5785-132	MCRI	Early Termination	3
5785-217	MCRI	Completed protocol	18
5785-256	MCRI	Completed protocol	18
5785-291	MCRI	Completed protocol	18
5785-325	MCRI	Completed protocol	18
5785-401	MCRI	Completed protocol	18
5785-404	MCRI	Early Termination	8
8644-18	SCHN	Early Termination	12
8644-22	SCHN	Completed protocol	13
8644-28	SCHN	Early Termination	13
8644-49	SCHN	Completed protocol	18
8644-53	SCHN	Completed protocol	18
8644-59	SCHN	Completed protocol	18

Table 10: Number and percentage of infants with scheduled and unscheduled skin prick tests.

Characteristic	aP	wP	Overall
	N = 486	N = 486	N = 972
Scheduled SPT	475 (98%)	473 (97%)	948 (98%)
Unscheduled SPTs			
0	462 (95%)	455 (94%)	917 (94%)
1	19 (3.9%)	29 (6.0%)	48 (4.9%)
2	5 (1.0%)	2 (0.4%)	7 (0.7%)

For the outcomes, a requirement was positive SPT by 18 months of age. Age of SPT in months was derived from `birthdat` (date of birth) and `pridat` (date of skin prick test) or in some cases `univdat` (date of unscheduled visit) when `pridat` was missing.

Most SPTs were performed by 13 months of age, although some were delayed, with a few occurring after 18-months of age (Figure 2). There were 5 infants who did have an SPT, but none by 18-months of age (Table 11). Participants who were enrolled under earlier stage 1 protocol versions had their SPT scheduled for 18 months of age rather than 12 months of age. Delays meant some of these SPT's did not occur until 19 or 20 months of age. For the outcomes, “approximately 12 months of age” was taken to mean “by 18 months of age” which then currently excludes some of these later SPTs.

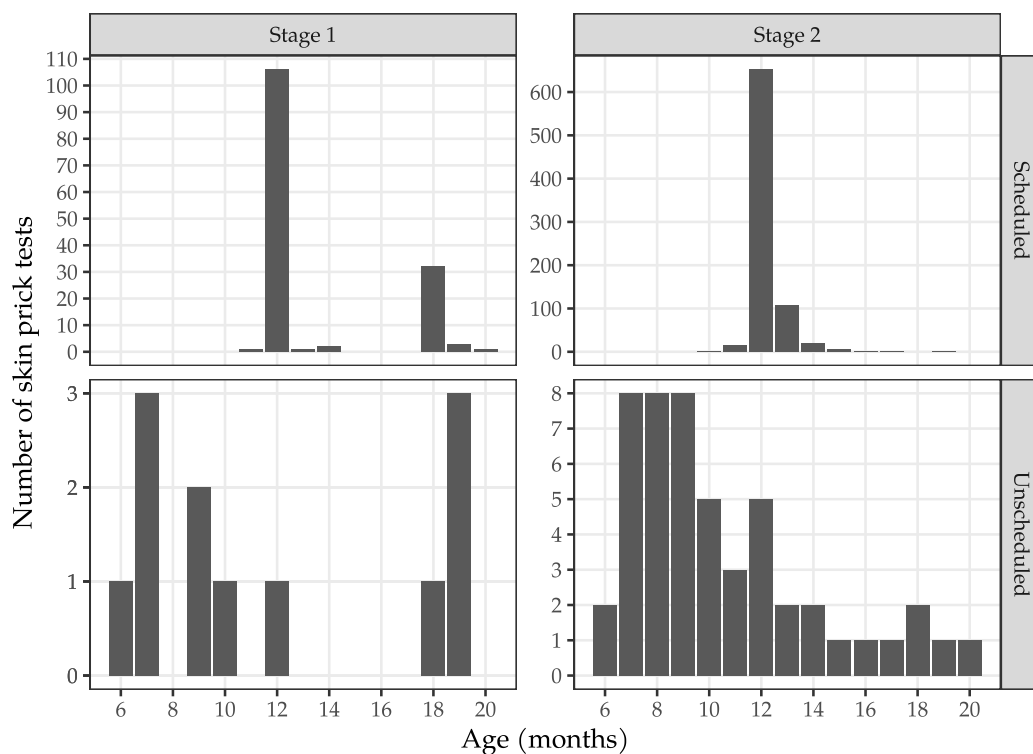


Figure 2: Age at skin prick test by protocol stage and occasion type (scheduled or unscheduled). Y-axis scale varies by panel.

Table 11: Listing of SPTs after 18-months of age.

Record ID	SPT type	SPT occasion	Age (months)
3	Scheduled	1	19
17	Unscheduled	2	19
25	Scheduled	1	19
34	Scheduled	1	19
36	Scheduled	1	20
68	Unscheduled	2	19
4629-17	Unscheduled	2	19
5785-10	Scheduled	1	19
5785-283	Unscheduled	2	20

Table 12 reports on the number and percentage of infants who had a SPT result of $>1\text{mm}$ and $\geq 3\text{mm}$ relative to negative control for at least one tested allergen (outcomes 2c and 2d). Any infants with no SPT records by 18 months of age are reported as missing.

Table 13 provides a breakdown of the specific allergens from the standard panel which satisfied the sensitisation criteria. Table 14 reports on any other tested allergens beyond the standard panel.

Whole egg has a higher proportion of missingness than other allergens from the standard panel, but this was usually because either egg white or egg yolk specifically was assessed and reported in the additional allergens instead of whole egg.

Table 12: Number and percentage of infants with a positive SPT by 18 months of age. Infants with no SPT records counted as missing.

Outcome	aP	wP	Overall
	N = 486	N = 486	N = 972
Any SPT $>1\text{mm}$ vs. neg control	81 (17%)	86 (18%)	167 (18%)
Missing	13	15	28
Any SPT $\geq 3\text{mm}$ vs neg. control	58 (12%)	60 (13%)	118 (13%)
Missing	13	15	28

Table 13: Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen (for those included in the standard panel). Each cell gives positive/tested (%) [missing result%].

Allergen	aP N = 486	wP N = 486	Overall N = 972
Wheal size >0mm			
D.pteronyssinus	19/474 (4.0%) [2.5%]	17/472 (3.6%) [2.9%]	36/946 (3.8%) [2.7%]
cat dander	11/474 (2.3%) [2.5%]	16/473 (3.4%) [2.7%]	27/947 (2.9%) [2.6%]
perennial ryegrass	10/474 (2.1%) [2.5%]	9/473 (1.9%) [2.7%]	19/947 (2.0%) [2.6%]
whole egg	41/458 (9.0%) [5.8%]	47/453 (10%) [6.8%]	88/911 (9.7%) [6.3%]
cashew	16/474 (3.4%) [2.5%]	16/472 (3.4%) [2.9%]	32/946 (3.4%) [2.7%]
cow's milk	18/472 (3.8%) [2.9%]	16/473 (3.4%) [2.7%]	34/945 (3.6%) [2.8%]
peanut	23/474 (4.9%) [2.5%]	24/471 (5.1%) [3.1%]	47/945 (5.0%) [2.8%]
Wheal size >1mm vs. neg control			
D.pteronyssinus	14/473 (3.0%) [2.7%]	15/472 (3.2%) [2.9%]	29/945 (3.1%) [2.8%]
cat dander	9/473 (1.9%) [2.7%]	10/473 (2.1%) [2.7%]	19/946 (2.0%) [2.7%]
perennial ryegrass	6/473 (1.3%) [2.7%]	6/473 (1.3%) [2.7%]	12/946 (1.3%) [2.7%]
whole egg	40/457 (8.8%) [6.0%]	44/453 (9.7%) [6.8%]	84/910 (9.2%) [6.4%]
cashew	14/473 (3.0%) [2.7%]	12/472 (2.5%) [2.9%]	26/945 (2.8%) [2.8%]
cow's milk	17/471 (3.6%) [3.1%]	15/473 (3.2%) [2.7%]	32/944 (3.4%) [2.9%]
peanut	20/473 (4.2%) [2.7%]	20/471 (4.2%) [3.1%]	40/944 (4.2%) [2.9%]
Wheal size ≥3mm vs neg. control			
D.pteronyssinus	10/473 (2.1%) [2.7%]	13/472 (2.8%) [2.9%]	23/945 (2.4%) [2.8%]
cat dander	2/473 (0.4%) [2.7%]	5/473 (1.1%) [2.7%]	7/946 (0.7%) [2.7%]
perennial ryegrass	3/473 (0.6%) [2.7%]	2/473 (0.4%) [2.7%]	5/946 (0.5%) [2.7%]
whole egg	29/457 (6.3%) [6.0%]	31/453 (6.8%) [6.8%]	60/910 (6.6%) [6.4%]
cashew	11/473 (2.3%) [2.7%]	8/472 (1.7%) [2.9%]	19/945 (2.0%) [2.8%]
cow's milk	14/471 (3.0%) [3.1%]	10/473 (2.1%) [2.7%]	24/944 (2.5%) [2.9%]
peanut	12/473 (2.5%) [2.7%]	13/471 (2.8%) [3.1%]	25/944 (2.6%) [2.9%]

Table 14: Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen for additional allergens which were tested. Each cell gives positive/tested (%). Totals restricted to only those infants with at least one additional allergen tested

Allergen	aP N = 86	wP N = 85	Overall N = 171
Wheal size >0mm			
sesame	1/64 (1.6%)	0/60 (0%)	1/124 (0.8%)
egg white	3/17 (18%)	6/21 (29%)	9/38 (24%)
egg yolk	4/17 (24%)	4/19 (21%)	8/36 (22%)
soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
banana	0/0 (NA%)	0/2 (0%)	0/2 (0%)
eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
dog	1/2 (50%)	0/0 (NA%)	1/2 (50%)
kiwi fruit	0/2 (0%)	0/0 (NA%)	0/2 (0%)
pork meat	0/0 (NA%)	0/1 (0%)	0/1 (0%)

sesame seed	0/0 (NA%)	0/1 (0%)	0/1 (0%)
tahini	0/0 (NA%)	0/1 (0%)	0/1 (0%)
wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)
whole wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)
pinenut	0/1 (0%)	0/0 (NA%)	0/1 (0%)
onion	0/1 (0%)	0/0 (NA%)	0/1 (0%)
garlic	0/1 (0%)	0/0 (NA%)	0/1 (0%)
raspberry	0/1 (0%)	0/0 (NA%)	0/1 (0%)
mango	0/1 (0%)	0/0 (NA%)	0/1 (0%)
ovomucoid	0/0 (NA%)	0/1 (0%)	0/1 (0%)
shrimp	1/1 (100%)	0/0 (NA%)	1/1 (100%)
tomato	0/0 (NA%)	0/1 (0%)	0/1 (0%)
d. farinae	0/0 (NA%)	0/1 (0%)	0/1 (0%)

Wheal size >1mm vs. neg control

sesame	1/64 (1.6%)	0/60 (0%)	1/124 (0.8%)
egg white	3/17 (18%)	4/21 (19%)	7/38 (18%)
egg yolk	4/17 (24%)	3/19 (16%)	7/36 (19%)
soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
banana	0/0 (NA%)	0/2 (0%)	0/2 (0%)
eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
dog	1/2 (50%)	0/0 (NA%)	1/2 (50%)
kiwi fruit	0/2 (0%)	0/0 (NA%)	0/2 (0%)
pork meat	0/0 (NA%)	0/1 (0%)	0/1 (0%)
sesame seed	0/0 (NA%)	0/1 (0%)	0/1 (0%)
tahini	0/0 (NA%)	0/1 (0%)	0/1 (0%)
wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)
whole wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)
pinenut	0/1 (0%)	0/0 (NA%)	0/1 (0%)
onion	0/1 (0%)	0/0 (NA%)	0/1 (0%)
garlic	0/1 (0%)	0/0 (NA%)	0/1 (0%)
raspberry	0/1 (0%)	0/0 (NA%)	0/1 (0%)
mango	0/1 (0%)	0/0 (NA%)	0/1 (0%)
ovomucoid	0/0 (NA%)	0/1 (0%)	0/1 (0%)
shrimp	1/1 (100%)	0/0 (NA%)	1/1 (100%)
tomato	0/0 (NA%)	0/1 (0%)	0/1 (0%)
d. farinae	0/0 (NA%)	0/1 (0%)	0/1 (0%)

Wheal size ≥3mm vs neg. control

sesame	1/64 (1.6%)	0/60 (0%)	1/124 (0.8%)
egg white	3/17 (18%)	3/21 (14%)	6/38 (16%)
egg yolk	2/17 (12%)	0/19 (0%)	2/36 (5.6%)
soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
banana	0/0 (NA%)	0/2 (0%)	0/2 (0%)
eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
dog	1/2 (50%)	0/0 (NA%)	1/2 (50%)
kiwi fruit	0/2 (0%)	0/0 (NA%)	0/2 (0%)
pork meat	0/0 (NA%)	0/1 (0%)	0/1 (0%)
sesame seed	0/0 (NA%)	0/1 (0%)	0/1 (0%)
tahini	0/0 (NA%)	0/1 (0%)	0/1 (0%)
wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)

whole wheat	0/1 (0%)	0/0 (NA%)	0/1 (0%)
pinenut	0/1 (0%)	0/0 (NA%)	0/1 (0%)
onion	0/1 (0%)	0/0 (NA%)	0/1 (0%)
garlic	0/1 (0%)	0/0 (NA%)	0/1 (0%)
raspberry	0/1 (0%)	0/0 (NA%)	0/1 (0%)
mango	0/1 (0%)	0/0 (NA%)	0/1 (0%)
ovomucoid	0/0 (NA%)	0/1 (0%)	0/1 (0%)
shrimp	1/1 (100%)	0/0 (NA%)	1/1 (100%)
tomato	0/0 (NA%)	0/1 (0%)	0/1 (0%)
d. farinae	0/0 (NA%)	0/1 (0%)	0/1 (0%)

3.1.1 Skin prick test model

Table 15 summarises the estimated model parameters for the probability of skin prick test sensitisation under each treatment group, and their difference (the posterior densities are shown in Figure 3 and Figure 4).

The results between the primary analysis (unadjusted) and secondary (adjusted) analyses were comparable. Given the data, under the assumed model the most credible values (based on 95% equal-tailed credible interval) for the difference in skin prick test sensitisation at 1mm [2.c] between wP and aP were (-0.04, 0.06) and at 3mm [2.d] were (-0.04, 0.05). The probability of a reduction in skin prick test sensitisation under wP compared to aP was 0.33 and 0.41 respectively.

Table 15: Posterior summaries for probability and difference in probability of positive skin prick test by size, unadjusted primary model. Point indicates median.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted					
>0mm	93/473 (20%)	0.20 (0.16, 0.23)	97/471 (21%)	0.21 (0.17, 0.24)	0.01 (-0.04, 0.06), 0.36
>1mm vs ctrl. [2.c]	81/473 (17%)	0.17 (0.14, 0.21)	86/471 (18%)	0.18 (0.15, 0.22)	0.01 (-0.04, 0.06), 0.32
≥3mm vs ctrl. [2.d]	58/473 (12%)	0.12 (0.10, 0.16)	60/471 (13%)	0.13 (0.10, 0.16)	0.00 (-0.04, 0.05), 0.41
Adjusted					
>0mm	93/473 (20%)	0.20 (0.16, 0.23)	97/471 (21%)	0.21 (0.17, 0.24)	0.01 (-0.04, 0.06), 0.36
>1mm vs ctrl. [2.c]	81/473 (17%)	0.17 (0.14, 0.21)	86/471 (18%)	0.18 (0.15, 0.22)	0.01 (-0.04, 0.06), 0.34
≥3mm vs ctrl. [2.d]	58/473 (12%)	0.12 (0.10, 0.16)	60/471 (13%)	0.13 (0.10, 0.16)	0.00 (-0.04, 0.04), 0.45

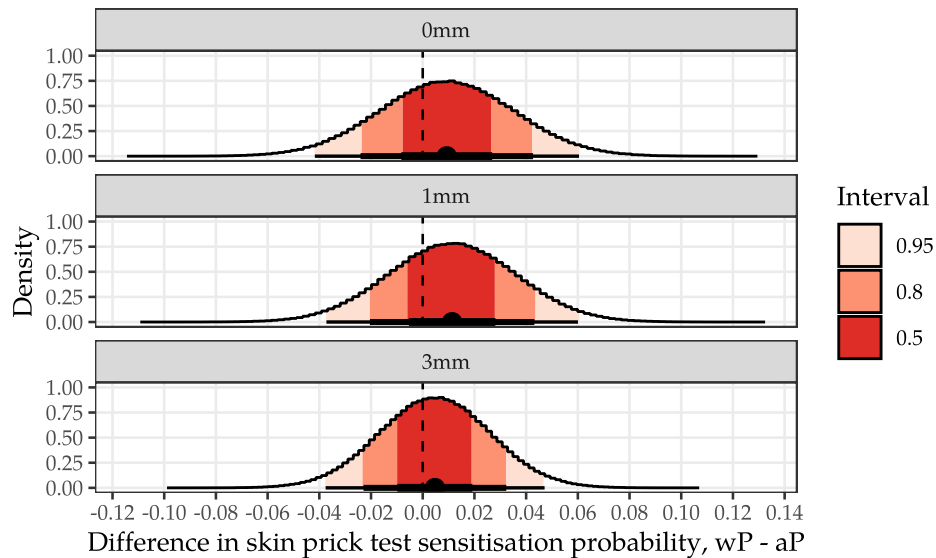


Figure 3: Posterior density for difference in probability of positive skin prick test by size, unadjusted primary model.

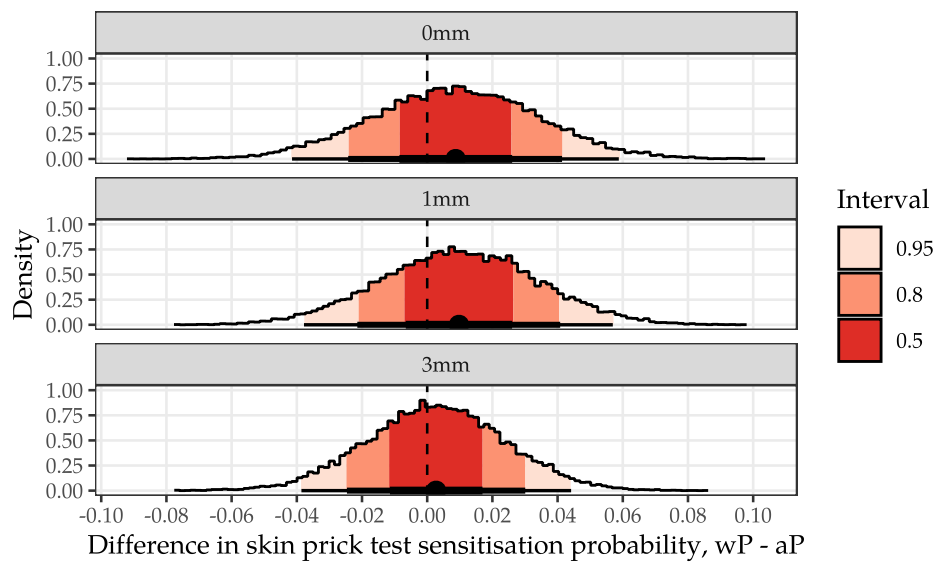


Figure 4: Posterior difference in probability of skin prick test sensitisation for wP compared to aP, adjusted model.

3.2 Food allergy

The primary source of information for food allergy was the `outcome_report` form. Reported outcomes were cross checked against the food and household questionnaire form (`food_and_household_questionnaire`) and the oral food challenge form (`food_challenge`).

Age at first reaction was derived from `outageval`. If this was missing then `birthdat` (date of birth) and `outdiagdat` (clinician-diagnosis date) was used to derive age in months.

In some cases participants may have withdrawn from the study without sufficient follow-up for food allergy to have been identified. The report makes no allowance for “missing” food allergy status, with absence of clinician diagnosed allergy (with or without adequate follow-up) counted as not meeting the primary outcome. Only 17 participants had an age of last contact 15 months or younger and may have been those most likely to have insufficient follow-up.

The following sections provide a summary of the number of infants who met each criteria. These infants are line listed in the appendix.

3.2.1 Oral food challenge

There were 31 outcome records for food allergy diagnoses resulting from oral food challenge amongst 23 infants. One reaction (record ID 8644-34) reportedly first occurred after 18 months of age and all others were by 18 months of age. All records had a matching positive skin prick test to the reported allergen and so all 22 infants met the primary outcome according to the definition (1.i).

3.2.2 Anaphylaxis

There were 6 outcome records for food allergy diagnoses resulting from anaphylaxis amongst 6 infants. Two of these infants (`record_id`: 68, 5785-283) were diagnosed after 18 months of age with first reported reaction occurring after 18 months of age. One infant (`record_id` = 8644-34) had no skin prick test for the reported allergens. So of the infants with anaphylaxis reported, 3 did not meet the criteria for the primary outcome according to the definition (1.ii).

3.2.3 Highly probable IgE mediated food allergy

There were 55 outcome records for food allergy diagnoses resulting from highly probable IgE mediated allergy amongst 43 infants. For one infant (`record_id`: 68), the age at first reaction and age at diagnosis were after 18 months of age. Four infants (`record_id`: 75, 5785-239, 5785-243, 5785-331) had no positive SPT to the reported allergen. Of the infants with IgE-mediated food allergy reported, 5 did not meet the primary outcome according to the definition (2.).

3.2.4 Primary outcome

Table 16 reports on the number of participants who met each of the primary outcome criteria (1.i, 1.ii, and 2) as outlined in the sections above.

Table 17 reports the total number of infants who met the primary outcome, and the primary reason grouped hierarchically according to the ordering in the definition (i.e. OFC → Anaphylaxis → IgE mediated) (irrespective of their timing).

Table 16: Number and percentage of infants who met each primary outcome condition and treatment group.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Met condition 1.i.	13 (2.7%)	9 (1.9%)	22 (2.3%)
Met condition 1.ii.	2 (0.4%)	1 (0.2%)	3 (0.3%)
Met condition 2.	17 (3.5%)	25 (5.1%)	42 (4.3%)

Table 17: Number and percentage of infants who met the primary outcome by primary reason and treatment group.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Met primary outcome	29 (6.0%)	32 (6.6%)	61 (6.3%)
Reason			
Criteria 1.i	13 (2.7%)	9 (1.9%)	22 (2.3%)
Criteria 1.ii	1 (0.2%)	1 (0.2%)	2 (0.2%)
Criteria 2.	15 (3.1%)	22 (4.5%)	37 (3.8%)

3.2.5 Primary outcome model

Table 18 summarises the estimated model parameters for the probability of food allergy under each treatment group, and their difference (the posterior densities are shown in Figure 5 and Figure 6).

The results between the primary analysis (unadjusted) and secondary (adjusted) analyses were comparable. Given the data, under the assumed model the most credible values (based on 95% equal-tailed credible interval) for the difference in food allergy between wP and aP were (-0.02, 0.04). The probability of a reduction in food allergy under wP compared to aP was 0.35.

Table 18: Posterior summaries for probability and difference in probability of food allergy, unadjusted primary model.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted	29/486 (6%)	0.06 (0.04, 0.08)	32/486 (7%)	0.07 (0.05, 0.09)	0.01 (-0.02, 0.04), 0.35
Adjusted	29/486 (6%)	0.06 (0.04, 0.08)	32/486 (7%)	0.07 (0.05, 0.09)	0.01 (-0.02, 0.04), 0.35

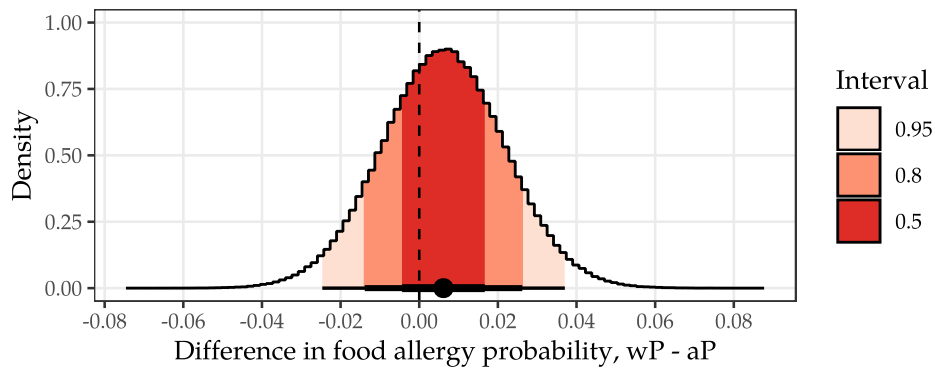


Figure 5: Posterior density for difference in probability of primary outcome, unadjusted primary model.

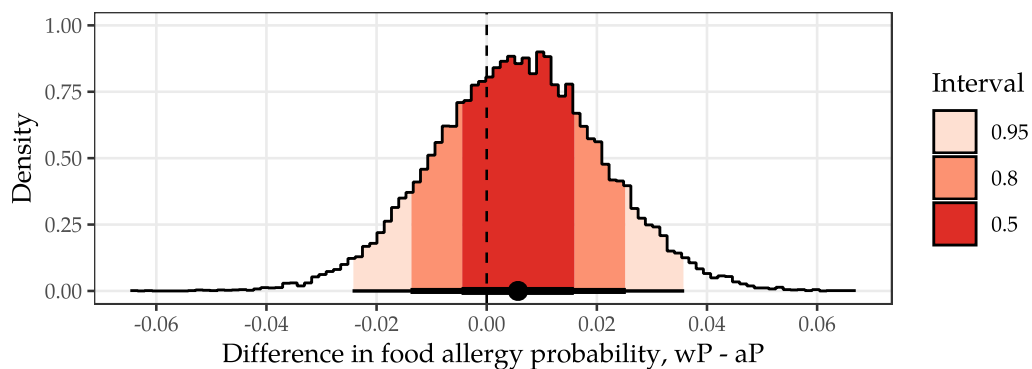


Figure 6: Posterior difference in probability of food allergy for wP compared to aP, adjusted model.

3.3 Eczema

The primary source of information for eczema was the `outcome_report` form. Reported outcomes were cross checked against the food and household questionnaire form and medical history forms. Infants whose first reaction date for clinician-diagnosed eczema was prior to enrolment were considered to have “pre-existing” eczema and were excluded from the analysis set. There were 8 such infants.

The number and percentage of infants with any clinician-diagnosed eczema is reported in Table 19. For the protocol defined outcomes [2a. and 2b.], the eczema diagnosis needed to be new-onset and accompanied by a positive SPT to at least one allergen by approximately 12 months of age. These are reported in Table 20.

Table 19: Number and percentage of infants with any clinician-diagnosed eczema.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Eczema	164 (34%)	158 (33%)	322 (33%)
Eczema by 6-months	89 (18%)	84 (17%)	173 (18%)
Eczema by 12-months	147 (30%)	135 (28%)	282 (29%)
Eczema by 18-months	163 (34%)	158 (33%)	321 (33%)

Table 20: Number and percentage of infants with new-onset clinician-diagnosed eczema. To be new-onset the first reaction date needed to be after the infants randomisation date. Those with missing were due to eczema diagnosis being present but no SPT records being available.

Outcome	aP N = 484	wP N = 480	Overall N = 964
New-onset cases			
Eczema	162 (33%)	152 (32%)	314 (33%)
Eczema by 6-months	87 (18%)	78 (16%)	165 (17%)
Eczema by 12-months	145 (30%)	129 (27%)	274 (28%)
Eczema by 18-months	161 (33%)	152 (32%)	313 (32%)
New-onset cases with positive SPT			
Eczema	50 (10%)	48 (10%)	98 (10%)
Missing	2	2	4
Eczema by 6-months	34 (7.0%)	28 (5.9%)	62 (6.5%)
Missing	1	2	3
Eczema by 12-months	47 (9.8%)	43 (9.0%)	90 (9.4%)
Missing	2	2	4
Eczema by 18-months	50 (10%)	48 (10%)	98 (10%)
Missing	2	2	4

3.3.1 Eczema model

Table 21 summarises the estimated model parameters for the probability of eczema by age under each treatment group, and their difference (the posterior densities are shown in Figure 7 and Figure 8).

The results between the primary analysis (unadjusted) and secondary (adjusted) analyses were comparable. Given the data, under the assumed primary model the most credible values (based on 95% equal-tailed credible interval) for the difference in eczema at 6 months [2.a] between wP and aP were (-0.04, 0.02) and at 12 months [2.b] (-0.04, 0.03). The probability of a reduction in eczema under wP compared to aP was 0.77 and 0.66 respectively.

Table 21: Posterior summaries for probability and difference in probability of eczema by age of onset, unadjusted primary model.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted					
6m [2.a]	34/483 (7%)	0.07 (0.05, 0.10)	28/478 (6%)	0.06 (0.04, 0.08)	-0.01 (-0.04, 0.02), 0.77
12m [2.b]	47/482 (10%)	0.10 (0.07, 0.13)	43/478 (9%)	0.09 (0.07, 0.12)	-0.01 (-0.04, 0.03), 0.66
18m	50/482 (10%)	0.10 (0.08, 0.13)	48/478 (10%)	0.10 (0.08, 0.13)	0.00 (-0.04, 0.04), 0.57
Adjusted					
6m [2.a]	34/483 (7%)	0.07 (0.05, 0.10)	28/478 (6%)	0.06 (0.04, 0.08)	-0.01 (-0.04, 0.02), 0.80
12m [2.b]	47/482 (10%)	0.10 (0.07, 0.13)	43/478 (9%)	0.09 (0.07, 0.12)	-0.01 (-0.05, 0.03), 0.68
18m	50/482 (10%)	0.11 (0.08, 0.13)	48/478 (10%)	0.10 (0.08, 0.13)	-0.01 (-0.04, 0.03), 0.62

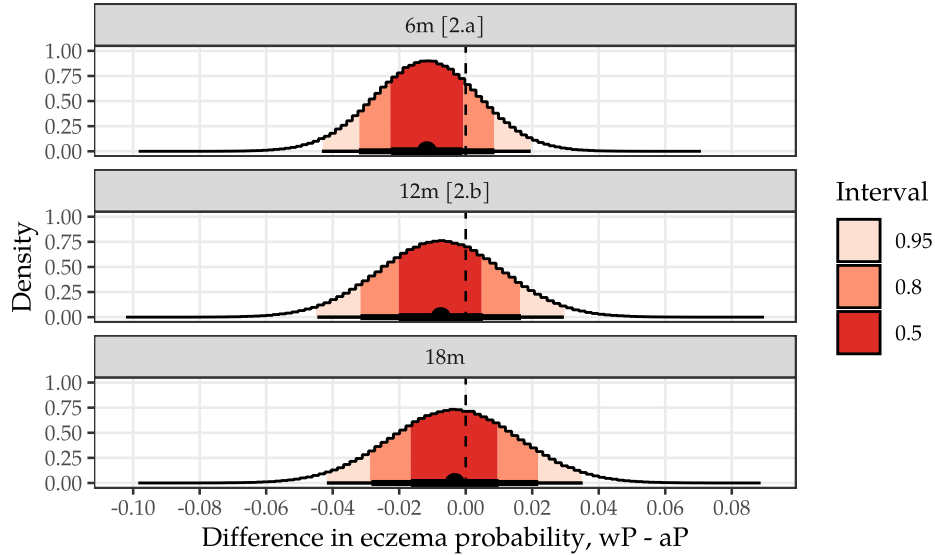


Figure 7: Posterior density for difference in probability of eczema diagnosis by age at reaction, unadjusted primary model.

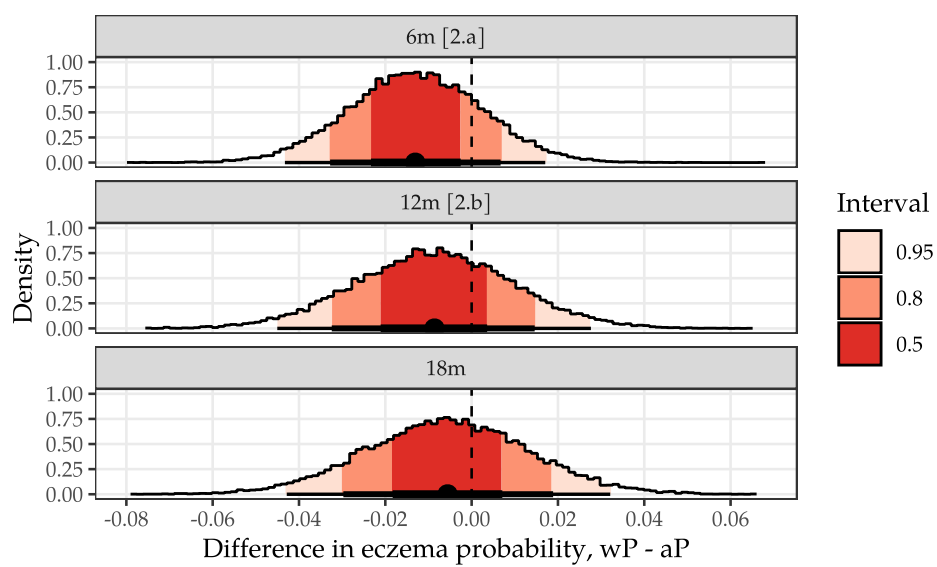


Figure 8: Posterior difference in probability of eczema diagnosis for wP compared to aP, adjusted model.

4 Safety

4.1 Serious adverse events

A listing of serious adverse events are reported in Table 22. These are any events reported either in the SAE reporting log (sae_reporting_log, stage 2 only) or on the adverse events form (adverse_events) with aetype = "SAE". In some cases, a record was on the SAE log but not the adverse event log.

Table 22: Listing of serious adverse events. Missing values were due to no adverse event record for corresponding SAE log record.

Record ID	Treatment	MedDRA code	Severity	Type of medical advice sought	Relationship to study vaccine?	Days after visit 1
22	wP	Bronchiolitis	Mild	Hospitalisation	Not Related	144
24	wP	Escherichia urinary tract infection	Moderate	Hospitalisation	Not Related	43
24	wP	Hospitalisation	Mild	Hospitalisation	Not Related	95
24	wP	Investigation	Mild	Hospitalisation	Not Related	190
60	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	49
65	aP	Hospitalisation	Mild	Hospitalisation	Not Related	7
94	wP	Rectal haemorrhage	Moderate	Hospitalisation	Not Related	7
4629-2	aP	Ileocolic intussusception	—	—	—	554
4629-5	aP	Laryngomalacia	—	—	—	48
4629-91	aP	Urinary tract infection	—	—	—	46
4629-158	aP	Gastroenteritis NOS	Moderate	Hospitalisation	Not Related	19
4629-175	wP	Irritability postvaccinal	Moderate	Hospitalisation	Related	0
4629-235	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	191
4629-235	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	209
4630-48	wP	Benign rolandic epilepsy	Moderate	Hospitalisation	Not Related	154
5785-140	wP	Gastroesophageal reflux	Severe	Hospitalisation	Not Related	3
8644-9	aP	Herniotomy	Moderate	Hospitalisation	Not Related	29
8644-10	wP	Acute bronchiolitis	Severe	Hospitalisation	Not Related	42

4.2 Adverse events

There were 183 adverse events documented during the trial. Most adverse events were for infants randomised at PCH (177) with only 6 adverse events for infants randomised at a site other than PCH (Table 23 and listing in Table 24).

There was a higher number of adverse events in the wP group compared to the aP group, but for both groups the majority of events were documented as unrelated to the blinded study vaccine (Table 25). A listing of the related AEs is given in Table 26.

A summary of the number and proportion of infants randomised at PCH with any reported adverse event by the MeDRA preferred term is given in Table 27. A summary of the total number of events is given in Table 28.

Table 23: Number of adverse events by study stage and site.

	aP (events/participants)	wP (events/participants)
Stage 1		
PCH	53/75	81/75
Stage 2		
PCH	14/131	29/132
CHW	2/30	1/29
MCRI	0/220	1/219
SCHN	1/30	1/31

Table 24: Adverse event listing of infants randomised at a site other than PCH.

Record ID	Site	Treatment	Preferred term	Severity	Relationship to study vaccine?	Type of medical advice sought
4630-19	CHW	aP	Pertussis	Moderate	Not Related	GP visit
4630-48	CHW	wP	Benign rolandic epilepsy	Moderate	Not Related	Hospitalisation
4630-55	CHW	aP	Respiratory syncytial virus infection	Severe	Not Related	Hospitalisation
5785-140	MCRI	wP	Gastroesophageal reflux	Severe	Not Related	Hospitalisation
8644-9	SCHN	aP	Herniotomy	Moderate	Not Related	Hospitalisation
8644-10	SCHN	wP	Acute bronchiolitis	Severe	Not Related	Hospitalisation

Table 25: Number and proportion of infants with any adverse events and the number of adverse events per infant, at PCH.

Characteristic	aP N = 206	wP N = 207	Overall N = 413
Any AE	49 (24%)	74 (36%)	123 (30%)
Any SAE	2 (1%)	6 (3%)	8 (2%)
Any related AE	2 (1%)	5 (2%)	7 (2%)
Any severe AE	3 (1%)	2 (1%)	5 (1%)

Number of AEs			
0	157 (76%)	133 (64%)	290 (70%)
1	36 (17%)	51 (25%)	87 (21%)
2	10 (5%)	16 (8%)	26 (6%)
3	2 (1%)	4 (2%)	6 (1%)
4	0 (0%)	2 (1%)	2 (0%)
5	1 (0%)	0 (0%)	1 (0%)
7	0 (0%)	1 (0%)	1 (0%)
Number of related AEs			
0	204 (99%)	202 (98%)	406 (98%)
1	1 (0%)	5 (2%)	6 (1%)
2	1 (0%)	0 (0%)	1 (0%)

Table 26: Listing of adverse events designated as “Related”.

Record ID	Treatment	Preferred term	Days after visit 1	Type of event	Severity	Relationship to study vaccine?	Type of medical advice sought
20	wP	Injection site haematoma	0	Unsolicited	Mild	Related	—
35	wP	Injection site haematoma	0	Unsolicited	Mild	Related	—
70	aP	Injection site haematoma	104	Unsolicited	Mild	Related	—
70	aP	Injection site haematoma	104	Unsolicited	Mild	Related	—
4629-75	wP	Rash trunk	1	Unsolicited	Mild	Related	—
4629-83	aP	Rash trunk	1	Unsolicited	Mild	Related	—
4629-101	wP	Injection site hematoma	—	Unsolicited	Mild	Related	—
4629-175	wP	Irritability postvaccinal	0	SAE	Moderate	Related	Hospitalisation

Table 27: Proportion of infants with any event by MedDRA preferred terms, PCH only.

Preferred term	aP N = 206	wP N = 207	Overall N = 413
Number of patients with event	49 (24%)	74 (36%)	123 (30%)
Abdominal distension	0 (0%)	1 (0.5%)	1 (0.2%)
Abdominal pain	0 (0%)	1 (0.5%)	1 (0.2%)
Adverse food reaction	1 (0.5%)	0 (0%)	1 (0.2%)
Arthropod bite	0 (0%)	1 (0.5%)	1 (0.2%)
Arthropod sting	0 (0%)	2 (1.0%)	2 (0.5%)
Balanoposthitis	0 (0%)	1 (0.5%)	1 (0.2%)
Bronchiolitis	0 (0%)	3 (1.4%)	3 (0.7%)
COVID-19	0 (0%)	1 (0.5%)	1 (0.2%)
Conjunctivitis	2 (1.0%)	2 (1.0%)	4 (1.0%)
Constipation	0 (0%)	3 (1.4%)	3 (0.7%)
Constipation neonatal	1 (0.5%)	0 (0%)	1 (0.2%)
Croup	0 (0%)	1 (0.5%)	1 (0.2%)
Dacryostenosis congenital	0 (0%)	1 (0.5%)	1 (0.2%)
Dermatitis	0 (0%)	1 (0.5%)	1 (0.2%)
Dermatitis contact	1 (0.5%)	0 (0%)	1 (0.2%)
Dermatitis diaper	0 (0%)	1 (0.5%)	1 (0.2%)
Dry skin	0 (0%)	2 (1.0%)	2 (0.5%)
Ear infection	1 (0.5%)	0 (0%)	1 (0.2%)
Eczema	12 (5.8%)	12 (5.8%)	24 (5.8%)

Escherichia urinary tract infection	0 (0%)	1 (0.5%)	1 (0.2%)
Flatulence	0 (0%)	1 (0.5%)	1 (0.2%)
Food allergy	2 (1.0%)	2 (1.0%)	4 (1.0%)
Gastroenteritis	1 (0.5%)	2 (1.0%)	3 (0.7%)
Gastroenteritis NOS	2 (1.0%)	0 (0%)	2 (0.5%)
Gastroenteritis viral	1 (0.5%)	1 (0.5%)	2 (0.5%)
Gastrooesophageal reflux disease	1 (0.5%)	2 (1.0%)	3 (0.7%)
Gingival pain	1 (0.5%)	0 (0%)	1 (0.2%)
Hospitalisation	1 (0.5%)	1 (0.5%)	2 (0.5%)
Inflammation at site of injection	0 (0%)	1 (0.5%)	1 (0.2%)
Injection site haematoma	3 (1.5%)	5 (2.4%)	8 (1.9%)
Injection site hematoma	0 (0%)	1 (0.5%)	1 (0.2%)
Investigation	0 (0%)	1 (0.5%)	1 (0.2%)
Irritability	1 (0.5%)	0 (0%)	1 (0.2%)
Irritability postvaccinal	0 (0%)	1 (0.5%)	1 (0.2%)
Lip injury	0 (0%)	1 (0.5%)	1 (0.2%)
Oral candidiasis	0 (0%)	1 (0.5%)	1 (0.2%)
Periorbital swelling	0 (0%)	1 (0.5%)	1 (0.2%)
Procedural pain	1 (0.5%)	0 (0%)	1 (0.2%)
Puffy eyes	1 (0.5%)	0 (0%)	1 (0.2%)
Rash	0 (0%)	5 (2.4%)	5 (1.2%)
Rash on leg	0 (0%)	1 (0.5%)	1 (0.2%)
Rash trunk	1 (0.5%)	1 (0.5%)	2 (0.5%)
Rectal haemorrhage	0 (0%)	1 (0.5%)	1 (0.2%)
Runny nose	0 (0%)	1 (0.5%)	1 (0.2%)
Scratch	1 (0.5%)	0 (0%)	1 (0.2%)
Seborrhoeic dermatitis	1 (0.5%)	0 (0%)	1 (0.2%)
Shaking of hands	0 (0%)	1 (0.5%)	1 (0.2%)
Sleep disorder	1 (0.5%)	0 (0%)	1 (0.2%)
Sore throat	0 (0%)	1 (0.5%)	1 (0.2%)
Teething	9 (4.4%)	12 (5.8%)	21 (5.1%)
Teething pain	1 (0.5%)	3 (1.4%)	4 (1.0%)
Tonsillitis	1 (0.5%)	1 (0.5%)	2 (0.5%)
Upper respiratory infection	0 (0%)	1 (0.5%)	1 (0.2%)
Upper respiratory tract infection	4 (1.9%)	7 (3.4%)	11 (2.7%)
Upper respiratory tract infection viral NOS	0 (0%)	1 (0.5%)	1 (0.2%)
Urticaria	1 (0.5%)	1 (0.5%)	2 (0.5%)
Vaccination site rash	1 (0.5%)	1 (0.5%)	2 (0.5%)
Vessel puncture site haematoma	1 (0.5%)	1 (0.5%)	2 (0.5%)
Vessel puncture site pain	1 (0.5%)	0 (0%)	1 (0.2%)
Viral infection	0 (0%)	1 (0.5%)	1 (0.2%)
Viral rash	0 (0%)	2 (1.0%)	2 (0.5%)
Viral upper respiratory tract infection	5 (2.4%)	7 (3.4%)	12 (2.9%)
Wheezing	1 (0.5%)	1 (0.5%)	2 (0.5%)

Table 28: Number of events by MedDRA preferred terms, PCH only.

Preferred term	aP N = 206	wP N = 207	Overall N = 413
Total number of events	67	110	177
Abdominal distension	0	2	2
Abdominal pain	0	1	1
Adverse food reaction	3	0	3
Arthropod bite	0	2	2
Arthropod sting	0	2	2
Balanoposthitis	0	1	1
Bronchiolitis	0	4	4
COVID-19	0	1	1
Conjunctivitis	2	2	4

Constipation	0	3	3
Constipation neonatal	1	0	1
Croup	0	1	1
Dacryostenosis congenital	0	1	1
Dermatitis	0	1	1
Dermatitis contact	1	0	1
Dermatitis diaper	0	1	1
Dry skin	0	2	2
Ear infection	1	0	1
Eczema	12	12	24
Escherichia urinary tract infection	0	1	1
Flatulence	0	1	1
Food allergy	2	2	4
Gastroenteritis	1	2	3
Gastroenteritis NOS	2	0	2
Gastroenteritis viral	1	1	2
Gastrooesophageal reflux disease	1	2	3
Gingival pain	1	0	1
Hospitalisation	1	1	2
Inflammation at site of injection	0	1	1
Injection site haematoma	4	5	9
Injection site hematoma	0	1	1
Investigation	0	1	1
Irritability	1	0	1
Irritability postvaccinal	0	1	1
Lip injury	0	1	1
Oral candidiasis	0	1	1
Periorbital swelling	0	1	1
Procedural pain	1	0	1
Puffy eyes	1	0	1
Rash	0	5	5
Rash on leg	0	1	1
Rash trunk	1	1	2
Rectal haemorrhage	0	1	1
Runny nose	0	1	1
Scratch	1	0	1
Seborrhoeic dermatitis	1	0	1
Shaking of hands	0	1	1
Sleep disorder	1	0	1
Sore throat	0	1	1
Teething	11	14	25
Teething pain	1	3	4
Tonsillitis	1	1	2
Upper respiratory infection	0	1	1
Upper respiratory tract infection	4	7	11
Upper respiratory tract infection viral NOS	0	1	1
Urticaria	1	1	2
Vaccination site rash	1	1	2
Vessel puncture site haematoma	1	1	2
Vessel puncture site pain	1	0	1
Viral infection	0	1	1
Viral rash	0	2	2
Viral upper respiratory tract infection	5	7	12
Wheezing	1	1	2

4.3 Diary Cards

Combined stage 1 and stage 2 diary card responses are reported in the following sections.

4.3.1 Returned

Table 29: Returned diary cards from stage 1.

Characteristic	aP N = 75	wP N = 75	Overall N = 150
6-week	72 (96%)	74 (99%)	146 (97%)
4-month	73 (97%)	71 (95%)	144 (96%)
6-month	68 (91%)	69 (92%)	137 (91%)
18-month	62 (83%)	62 (83%)	124 (83%)

Table 30: Returned diary cards from stage 2.

Characteristic	aP N = 77	wP N = 76	Overall N = 153
6-week	73 (95%)	63 (83%)	136 (89%)
18-month	47 (61%)	62 (82%)	109 (71%)

4.3.2 Solicited systemic reactions

4.3.2.1 Irritability

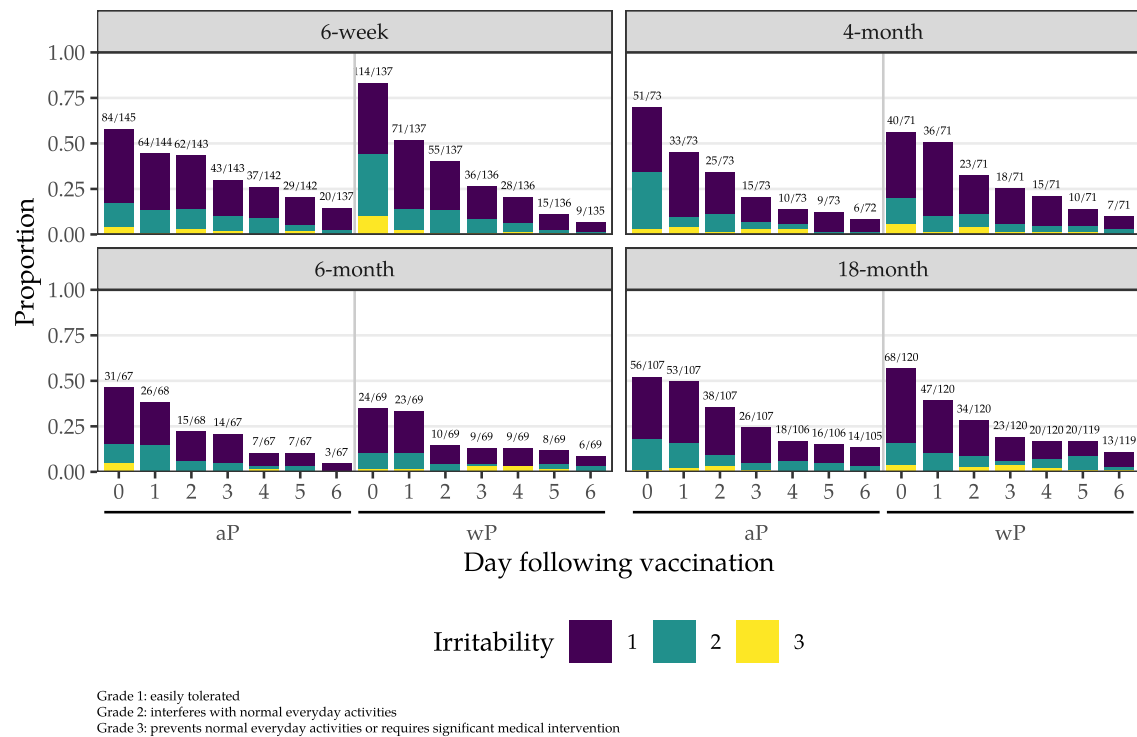


Figure 9: Irritability day 0 to 6 by vaccination age and assigned treatment.

4.3.2.2 Vomiting

Figure 10 presents the distribution of daily grades for vomiting following each vaccination occasion by assigned treatment.

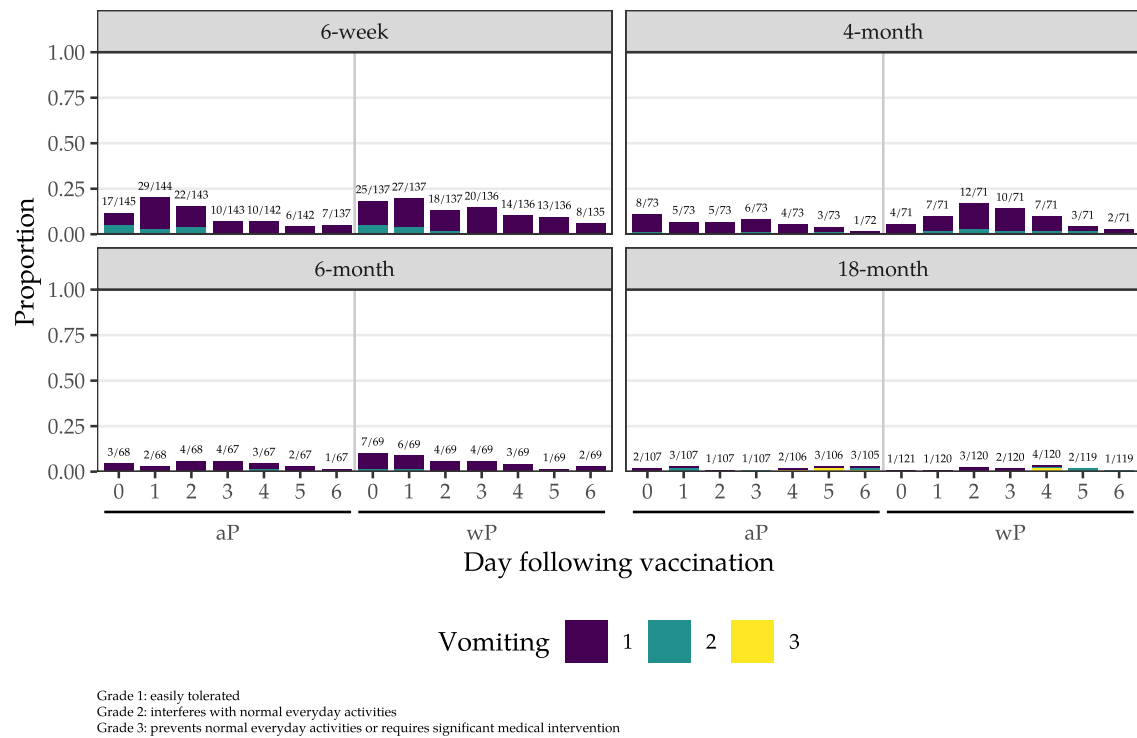


Figure 10: Vomiting day 0 to 6 by vaccination age and assigned treatment.

4.3.2.3 Diarrhoea

Figure 11 presents the distribution of daily grades for diarrhoea following each vaccination occasion by assigned treatment.

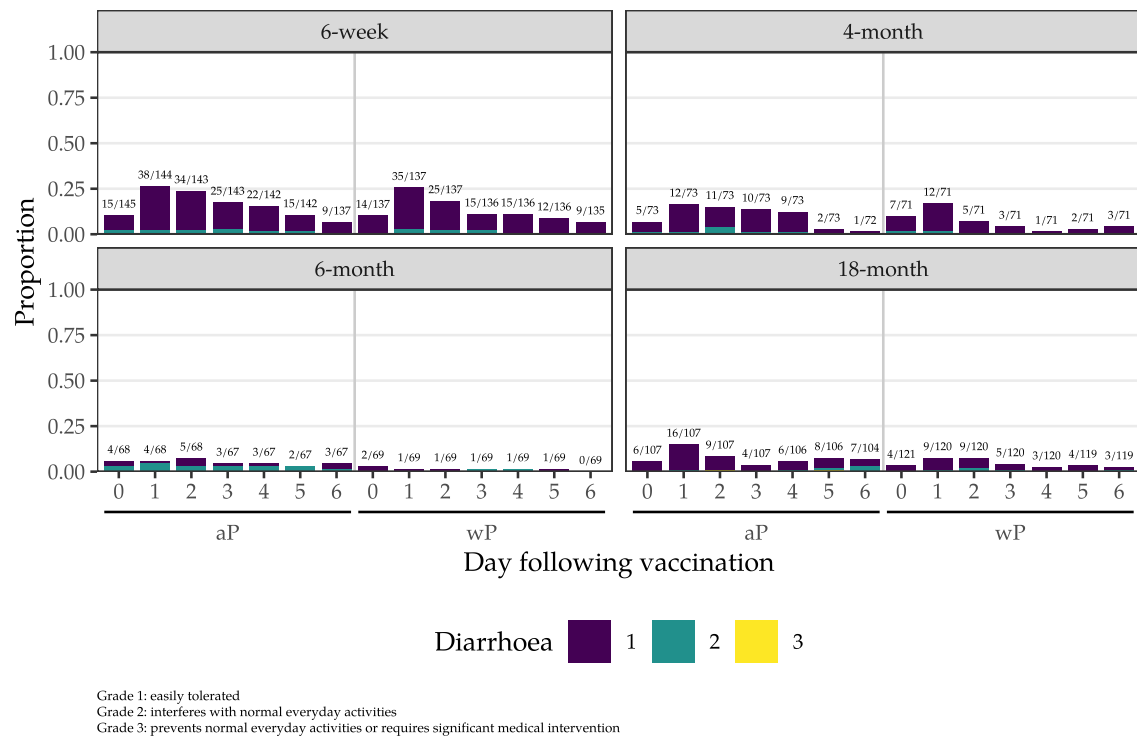


Figure 11: Diarrhoea day 0 to 6 by vaccination age and assigned treatment.

4.3.2.4 Decreased Feeding

Figure 12 presents the distribution of daily intensity grades for decreased feeding following each vaccination occasion by assigned treatment.

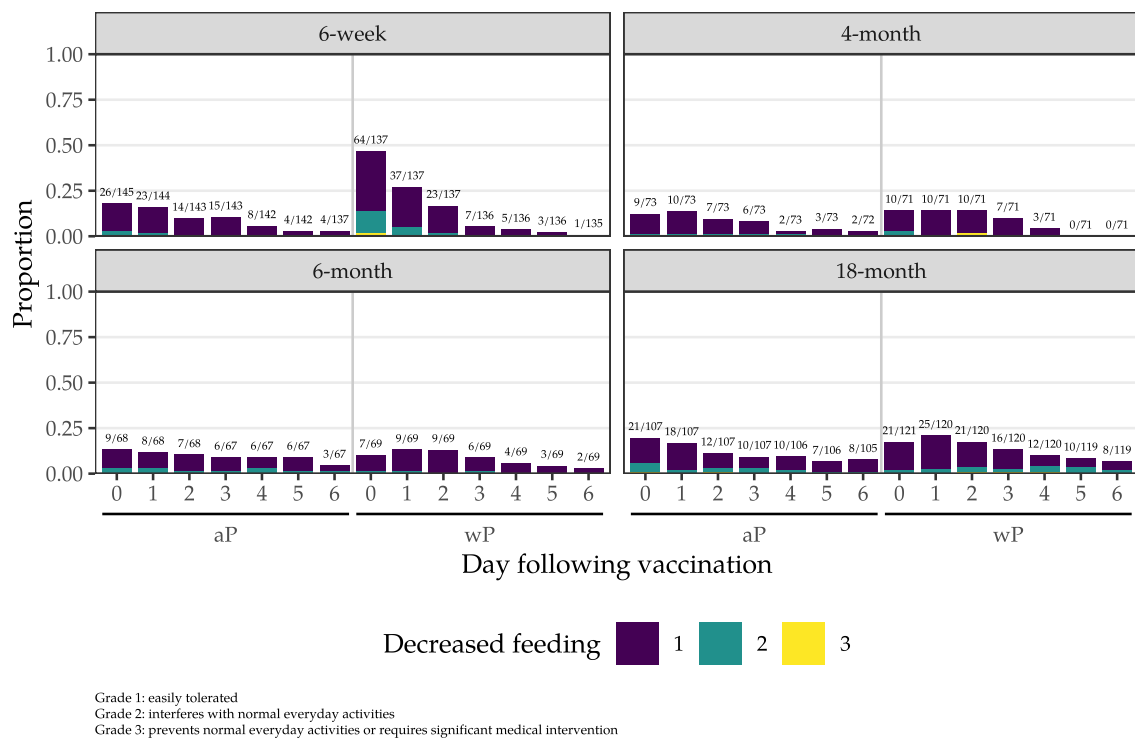


Figure 12: Decreased feeding day 0 to 6 by vaccination age and assigned treatment.

4.3.2.5 Drowsiness

Figure 13 presents the distribution of daily intensity grades for drowsiness following each vaccination occasion by assigned treatment.

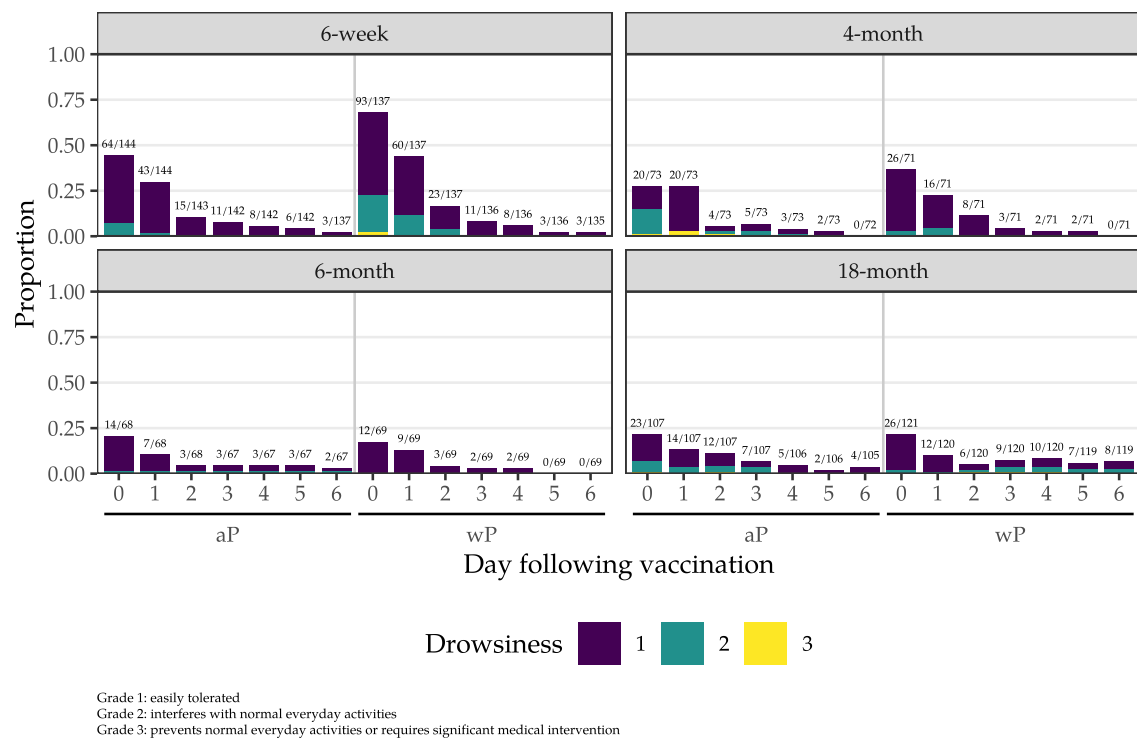


Figure 13: Drowsiness day 0 to 6 by vaccination age and assigned treatment.

4.3.2.6 Restlessness

Figure 14 presents the distribution of daily intensity grades for restlessness following each vaccination occasion by assigned treatment.

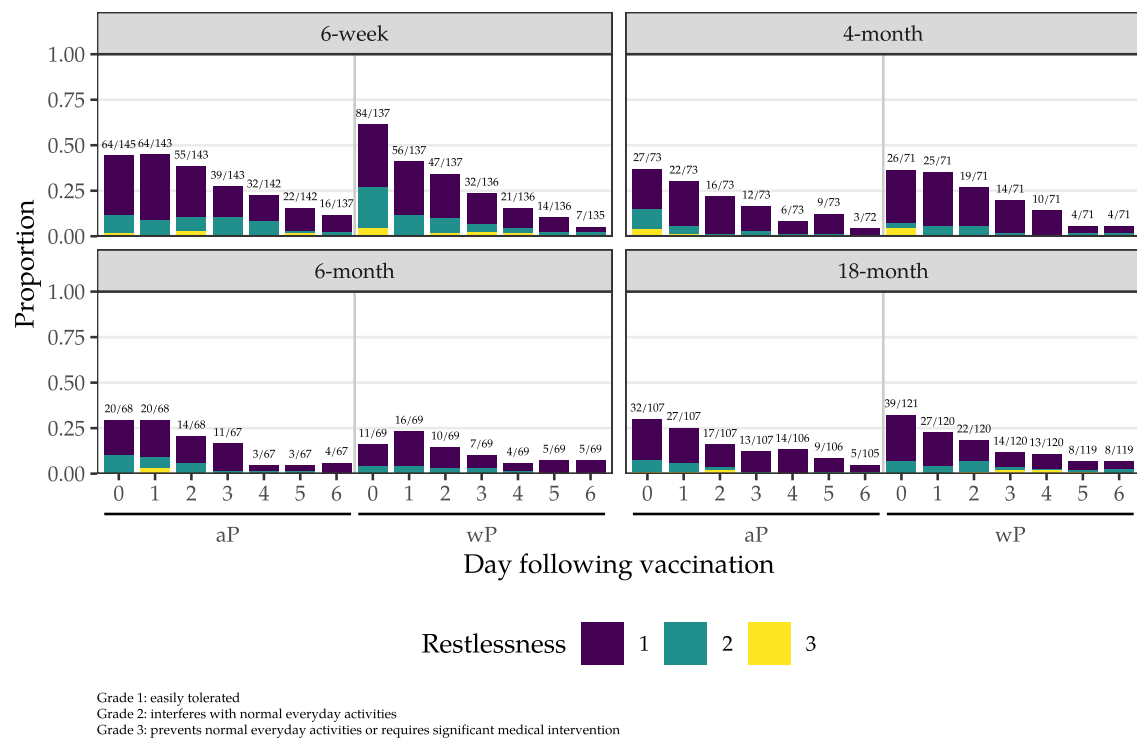


Figure 14: Restlessness day 0 to 6 by vaccination age and assigned treatment.

4.3.2.7 Fever

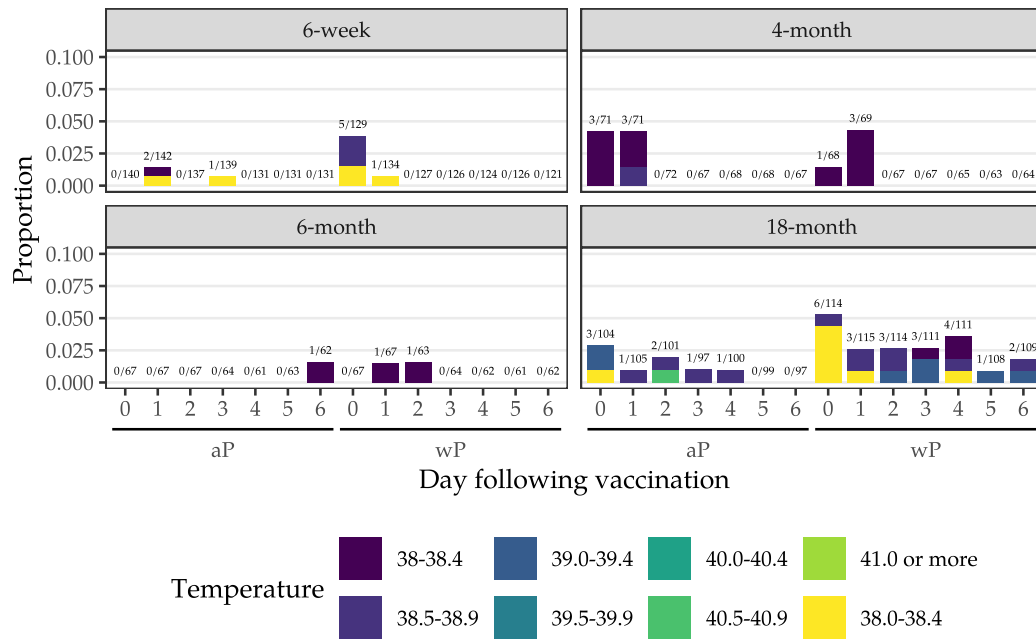


Figure 15: Fever by vaccination age and assigned treatment, stage 1 and 2 combined.

4.3.3 Solicited Injection Site Reactions

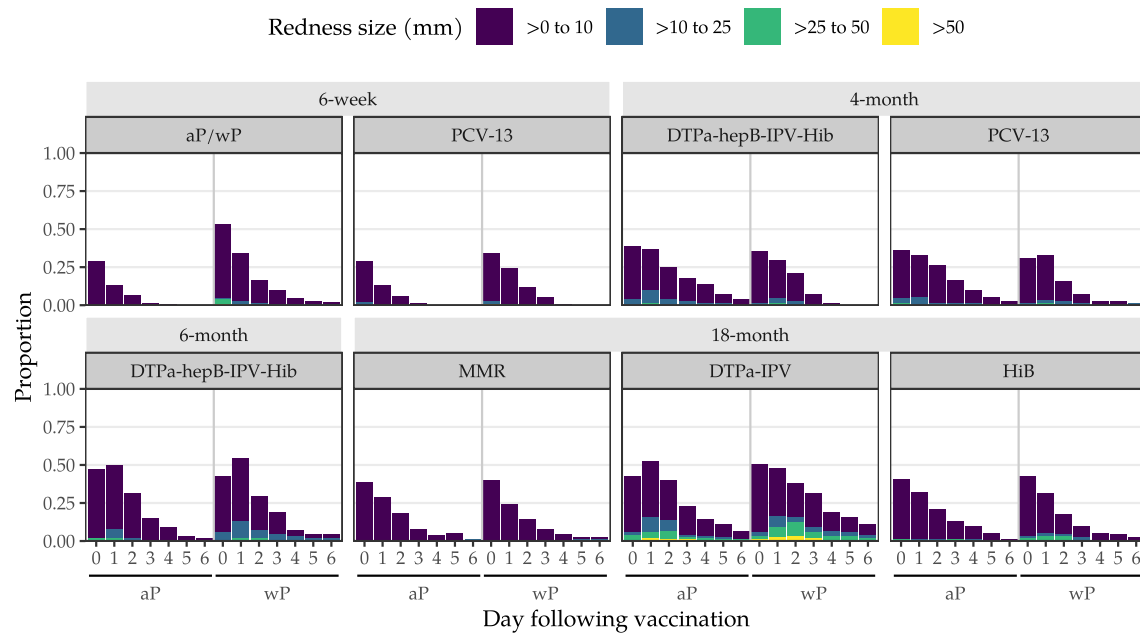


Figure 16: Injection site erythema by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.

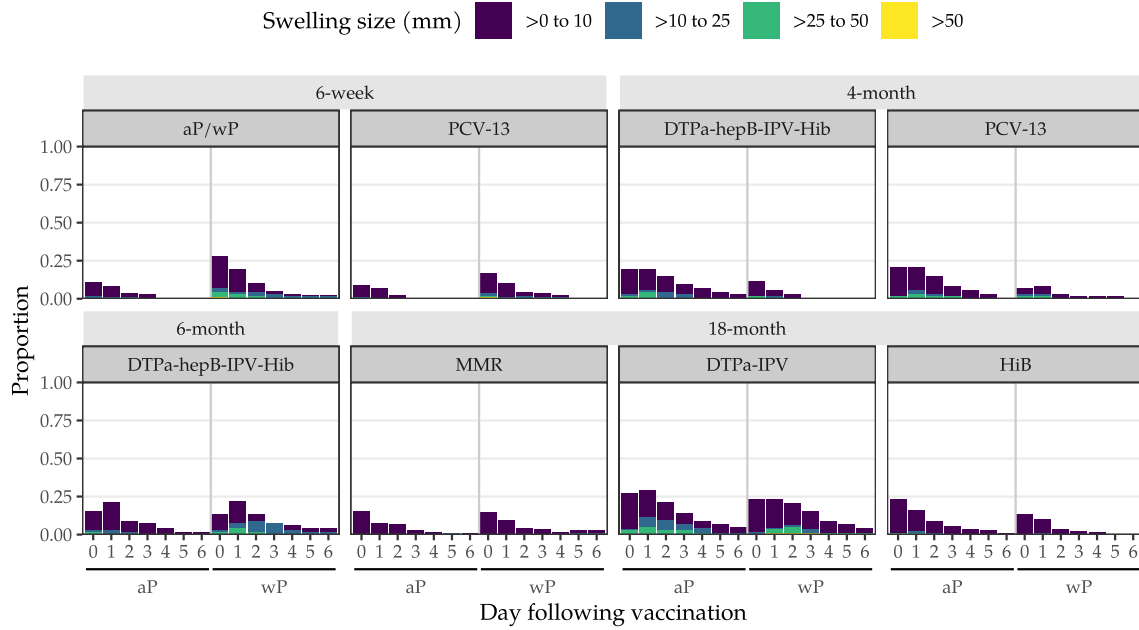


Figure 17: Injection site swelling by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.

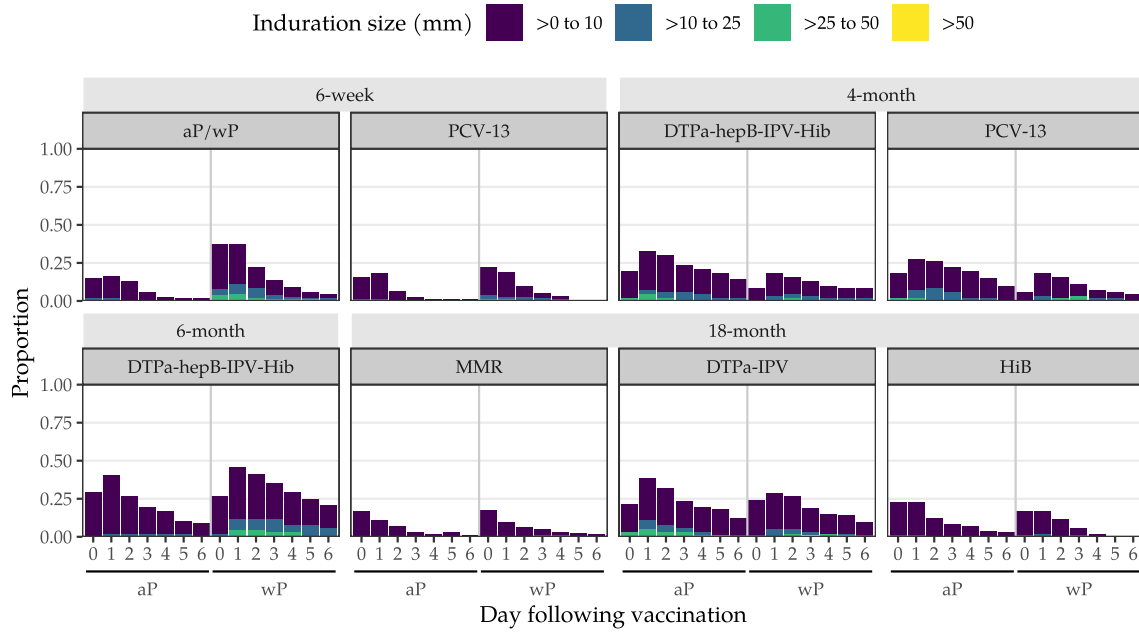


Figure 18: Injection site induration by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.

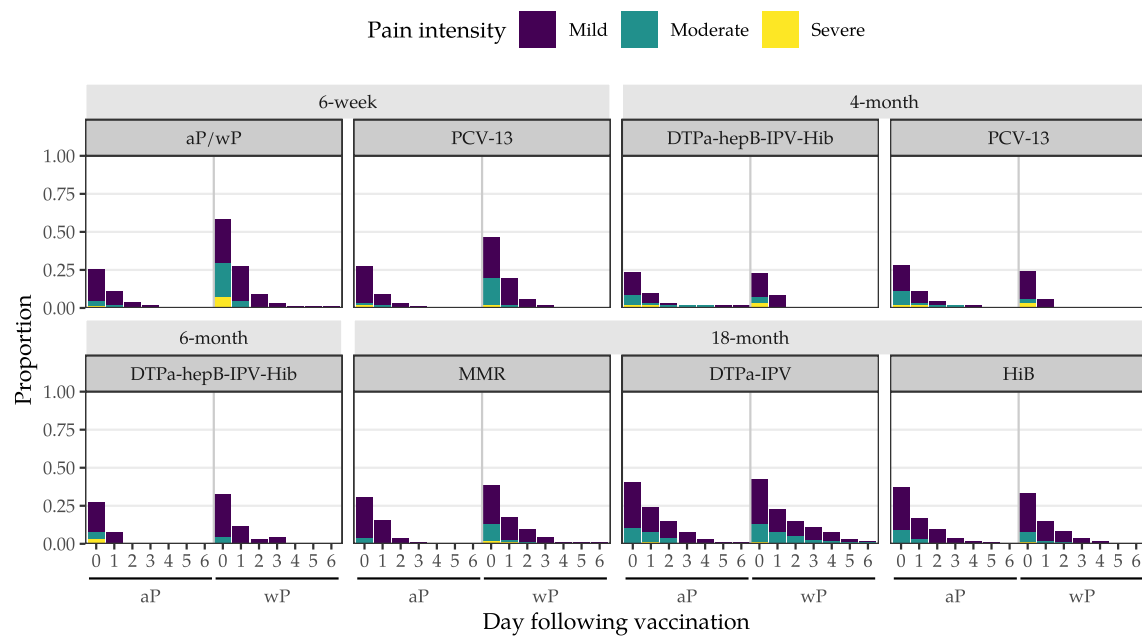


Figure 19: Injection site pain by vaccination age, vaccine, and assigned treatment.

5 Appendix

5.1 Primary outcome listings

Table 31: Listing of infants with oral food challenge outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	OFC positive foods	Foods with positive SPT
4629-32	wP	egg	egg
4629-228	aP	peanut	peanut
4630-30	aP	egg	egg
4630-45	wP	peanut	peanut
5785-39	aP	peanut, tahini	peanut
5785-78	aP	peanut, cashew, cooked egg	peanut, cashew, egg
5785-98	aP	whole egg cooked	whole
5785-112	aP	egg	egg
5785-122	aP	peanut	peanut
5785-123	aP	egg	egg
5785-133	aP	peanut, cows milk	peanut, milk
5785-169	aP	cows milk	milk
5785-192	wP	cow milk	milk
5785-242	wP	peanut, cows milk	peanut, milk
5785-266	aP	well cooked scrambled egg	egg
5785-295	aP	cows milk	milk
5785-319	wP	cashew	cashew
5785-320	aP	egg	egg
5785-333	wP	peanut, cows milk, egg	peanut, milk, egg
5785-367	wP	peanut, cooked egg	peanut, egg
8644-14	wP	hazelnut	hazelnut
8644-24	wP	cashew	cashew

Table 32: Listing of infants with anaphylaxis outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	Anaphylaxis foods	Foods with positive SPT
136	wP	cows milk	milk
4629-92	aP	cows milk	milk
5785-133	aP	scrambled egg	egg
8644-34	aP	miso and salmon	

Table 33: Listing of infants with IgE-mediated food allergy outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	IgE mediated foods	Foods with positive SPT
17	wP	peanut	peanut
21	aP	egg	egg
56	wP	egg	egg
73	wP	peanut	peanut
75	aP	egg	
115	wP	cows milk, peanut, egg	milk, peanut, egg
117	aP	cows milk	milk
4629-5	aP	cashews, almonds and coconut	cashew, almond, coconut
4629-60	wP	egg	egg
4629-87	aP	egg	egg
4629-92	aP	pecan, egg allergy	pecan, egg
4629-181	wP	egg	egg
4629-189	wP	egg	egg
4629-191	wP	peanut	peanut
4629-214	wP	egg	egg
4629-226	wP	egg, peanut	egg, peanut
4629-253	wP	egg powder	egg
4630-7	aP	soy, dairy, egg, beef	soy, egg, beef
4630-17	aP	egg	egg
4630-57	wP	egg	egg
5785-1	wP	dairy	milk
5785-7	wP	egg	egg
5785-22	aP	egg	egg
5785-89	wP	well cooked scrambled egg	egg
5785-93	wP	egg	egg
5785-94	aP	cooked egg	egg
5785-98	aP	cow milk formula	milk
5785-124	wP	cows milk	milk
5785-145	aP	egg	egg
5785-200	wP	cows milk	milk
5785-203	aP	egg, milk	egg, milk
5785-239	aP	steamed egg plant	
5785-242	wP	baked egg	egg
5785-243	aP	cows milk formula, rice cereal	
5785-301	wP	cashew, peanut	cashew, peanut
5785-331	wP	soft boiled egg	
5785-367	wP	peanut	peanut
5785-372	wP	peanut	peanut
5785-380	wP	egg	egg
5785-384	aP	egg	egg
8644-14	wP	whole egg, peanut	egg, peanut
8644-34	aP	sesame, peanut	sesame, peanut

5.2 Secondary outcome listings

Table 34: Listing of infants with eczema diagnosis and positive SPT to any allergen.

Record ID	Treatment	Age at first reaction (months)
17	wP	13
56	wP	8
68	aP	5
73	wP	4
115	wP	6
117	aP	12
149	wP	5
4629-21	aP	6
4629-31	wP	4
4629-32	wP	10
4629-33	aP	16
4629-39	wP	6
4629-49	wP	12
4629-60	wP	6
4629-61	aP	6
4629-87	aP	5
4629-92	aP	5
4629-117	aP	7
4629-132	aP	4
4629-145	aP	7
4629-168	wP	3
4629-191	wP	3
4629-205	wP	13
4629-212	aP	9
4629-214	wP	12
4629-215	wP	14
4629-228	aP	9
4629-239	wP	6
4629-245	aP	16
4629-253	wP	7
4629-260	aP	5
4630-3	aP	4
4630-4	aP	5
4630-7	aP	2
4630-14	wP	12
4630-17	aP	4
4630-20	aP	6
4630-22	aP	4
4630-23	aP	6
4630-27	aP	3
4630-28	wP	10
4630-32	wP	4
4630-33	aP	5
4630-43	aP	11
4630-45	wP	4
4630-54	wP	4
4630-57	wP	4
5785-1	wP	11
5785-2	wP	4
5785-7	wP	3
5785-22	aP	3
5785-39	aP	3
5785-44	wP	8
5785-49	aP	9
5785-53	wP	5

5785-70	wP	3
5785-78	aP	5
5785-91	wP	5
5785-93	wP	3
5785-98	aP	4
5785-112	aP	4
5785-122	aP	2
5785-123	aP	3
5785-124	wP	7
5785-133	aP	9
5785-145	aP	6
5785-169	aP	3
5785-170	wP	4
5785-192	wP	7
5785-203	aP	6
5785-207	wP	6
5785-211	aP	6
5785-218	aP	6
5785-265	aP	6
5785-266	aP	6
5785-275	aP	4
5785-283	wP	2
5785-288	aP	14
5785-295	aP	5
5785-301	wP	3
5785-319	wP	3
5785-320	aP	8
5785-332	wP	6
5785-349	aP	6
5785-358	wP	9
5785-370	aP	3
5785-372	wP	8
5785-380	wP	6
5785-384	aP	2
8644-14	wP	2
8644-19	aP	7
8644-24	wP	18
8644-27	wP	3
8644-34	aP	4
8644-41	aP	3
8644-44	wP	4
8644-51	wP	2
8644-58	wP	15

5.3 IgE responses

The 6 and 7 month IgE responses were previously reported. What follows additionally reports on 19-month IgE responses. Descriptive statistics are reported in Table 35.

Table 35: Sample IgE concentration summaries (kU/L) by allergen type, visit age, and assigned vaccination schedule.

	aP					wP				
	n	GMC ¹	Min–Max	≥0.01	≥0.1	n	GMC ¹	Min–Max	≥0.01	≥0.1
Total IgE										
6-month	67	4.76	0.40–121.43	100%	100%	61	6.11	0.19–141.38	100%	100%
7-month	66	6.36	0.52–241.99	100%	100%	57	7.71	0.52–148.70	100%	100%
19-month	62	17.64	0.92–254.03	100%	100%	53	20.43	1.87–951.55	100%	100%
Tetanus-toxoid										
6-month	67	0.00	0.00–0.22	12%	1%	61	0.00	0.00–1.71	21%	7%
7-month	66	0.00	0.00–0.20	21%	3%	57	0.00	0.00–0.82	21%	7%
19-month	65	0.02	0.00–4.39	65%	26%	55	0.02	0.00–5.05	64%	31%
Egg white										
6-month	67	0.00	0.00–14.91	36%	18%	61	0.00	0.00–3.00	26%	20%
7-month	67	0.01	0.00–39.50	54%	24%	57	0.00	0.00–4.51	42%	19%
19-month	64	0.03	0.00–59.70	75%	22%	54	0.04	0.00–6.67	74%	28%
Whole egg										
6-month	67	0.00	0.00–14.85	40%	18%	60	0.00	0.00–3.51	42%	22%
7-month	67	0.01	0.00–33.70	52%	22%	57	0.00	0.00–5.16	40%	21%
19-month	65	0.03	0.00–64.03	58%	26%	55	0.03	0.00–6.95	71%	27%
Cashew										
19-month	64	0.01	0.00–3.99	45%	6%	53	0.01	0.00–2.65	45%	9%
Cat										
19-month	60	0.01	0.00–3.02	37%	2%	47	0.01	0.00–1.29	34%	2%
Dust										
19-month	60	0.02	0.01–0.20	97%	3%	47	0.02	0.01–0.43	98%	2%
Cow's milk										
19-month	62	0.05	0.01–6.32	98%	21%	53	0.06	0.01–159.54	100%	21%
Peanut										
19-month	64	0.03	0.01–79.32	89%	12%	55	0.04	0.01–3.51	93%	16%
Rye										
19-month	58	0.01	0.00–0.12	52%	2%	47	0.01	0.00–1.68	45%	4%

¹Values of 0.00 kU/L and 0.000 kU/L set to 0.005 and 0.0005 respectively for calculation of sample GMC.

5.3.1 Geometric mean concentration

Table 36: Geometric mean ratio (GMR) and geometric mean fold rise ratio (GMFRR) for total IgE.

	Mean $\pm$ std	Median	95% CrI	Pr(< 1)
Total IgE				
6-month	1.01 $\pm$ 0.22	0.99	(0.65, 1.51)	0.53
7-month	0.95 $\pm$ 0.20	0.92	(0.61, 1.40)	0.65
19-month	0.97 $\pm$ 0.24	0.94	(0.59, 1.53)	0.60
7/6-month (GMFR)	0.94 $\pm$ 0.13	0.94	(0.72, 1.22)	0.69
19/7-month (GMFR)	1.03 $\pm$ 0.20	1.01	(0.70, 1.48)	0.47
Tetanus toxoid				
6-month	12.16 $\pm$ 36.35	4.94	(0.57, 63.58)	0.07
7-month	2.72 $\pm$ 3.70	1.76	(0.33, 10.98)	0.25
19-month	1.38 $\pm$ 0.63	1.26	(0.53, 2.91)	0.29
7/6-month (GMFR)	0.46 $\pm$ 0.45	0.35	(0.06, 1.50)	0.92
19/7-month (GMFR)	0.91 $\pm$ 0.75	0.72	(0.16, 2.80)	0.69
Egg white				
6-month	0.75 $\pm$ 0.88	0.50	(0.08, 3.00)	0.79
7-month	0.82 $\pm$ 0.64	0.64	(0.17, 2.48)	0.75
19-month	1.10 $\pm$ 0.39	1.05	(0.53, 2.02)	0.45
7/6-month (GMFR)	1.60 $\pm$ 1.27	1.31	(0.41, 4.62)	0.33
19/7-month (GMFR)	1.98 $\pm$ 1.47	1.62	(0.47, 5.69)	0.22
Whole egg				
6-month	1.27 $\pm$ 1.14	0.97	(0.22, 4.23)	0.52
7-month	0.92 $\pm$ 0.78	0.71	(0.18, 2.90)	0.69
19-month	1.28 $\pm$ 0.56	1.17	(0.52, 2.66)	0.35
7/6-month (GMFR)	0.82 $\pm$ 0.38	0.75	(0.31, 1.75)	0.75
19/7-month (GMFR)	2.05 $\pm$ 1.56	1.63	(0.45, 6.05)	0.22

5.3.2 Sensitisation

Table 37: Summary of sensitisation model by response type.

	Standardised Pr(IgE $\geq$ 0.1), Median (95% CrI)				Odds ratios	
	aP	wP	wP – aP	Pr(wP – aP < 0)	Conditional	Marginal
Tetanus-toxoid						
6-month	0.04 (0.01, 0.10)	0.05 (0.02, 0.12)	0.01 (-0.03, 0.07)	0.26	1.84 (0.27, 13.38)	1.49 (0.42, 6.62)
7-month	0.04 (0.01, 0.11)	0.06 (0.02, 0.13)	0.01 (-0.05, 0.09)	0.31	1.80 (0.14, 21.01)	1.49 (0.25, 9.48)
19-month	0.26 (0.16, 0.37)	0.32 (0.21, 0.45)	0.06 (-0.08, 0.21)	0.18	2.09 (0.39, 12.62)	1.45 (0.62, 3.45)
Egg white						
6-month	0.19 (0.10, 0.30)	0.20 (0.11, 0.32)	0.01 (-0.09, 0.12)	0.42	1.21 (0.18, 8.51)	1.07 (0.54, 2.18)
7-month	0.24 (0.15, 0.37)	0.22 (0.12, 0.34)	-0.03 (-0.15, 0.10)	0.68	0.61 (0.06, 5.42)	0.84 (0.37, 1.81)
19-month	0.25 (0.14, 0.36)	0.27 (0.16, 0.39)	0.02 (-0.11, 0.15)	0.38	1.40 (0.16, 12.55)	1.12 (0.53, 2.50)
Whole egg						
6-month	0.20 (0.11, 0.31)	0.22 (0.13, 0.34)	0.02 (-0.08, 0.13)	0.32	1.56 (0.21, 11.83)	1.15 (0.60, 2.31)
7-month	0.23 (0.13, 0.35)	0.24 (0.14, 0.37)	0.01 (-0.11, 0.14)	0.42	1.24 (0.13, 12.71)	1.07 (0.52, 2.28)
19-month	0.28 (0.17, 0.39)	0.27 (0.16, 0.39)	-0.01 (-0.13, 0.12)	0.56	0.84 (0.08, 7.92)	0.95 (0.44, 2.01)

5.3.3 Quantitation

Table 38: Summary of sensitisation model for tetanus toxoid IgE.

	Standardised Pr(IgE ≥ 0.1), Median (95% CrI)			Pr(wP – aP < 0)	Odds ratios	
	aP	wP	wP – aP		Conditional	Marginal
Tetanus-toxoid						
6-month	0.17 (0.09, 0.28)	0.20 (0.11, 0.32)	0.03 (-0.07, 0.13)	0.28	1.78 (0.25, 13.32)	1.22 (0.61, 2.56)
7-month	0.23 (0.13, 0.35)	0.22 (0.12, 0.34)	-0.01 (-0.13, 0.12)	0.56	0.84 (0.08, 8.69)	0.94 (0.41, 2.13)
19-month	0.62 (0.50, 0.73)	0.64 (0.51, 0.76)	0.03 (-0.11, 0.16)	0.35	1.54 (0.18, 14.82)	1.15 (0.56, 2.39)
Egg white						
6-month	0.36 (0.25, 0.48)	0.28 (0.17, 0.40)	-0.08 (-0.22, 0.05)	0.89	0.47 (0.13, 1.60)	0.66 (0.32, 1.29)
7-month	0.52 (0.39, 0.64)	0.44 (0.31, 0.57)	-0.08 (-0.23, 0.08)	0.84	0.52 (0.13, 1.93)	0.71 (0.34, 1.41)
19-month	0.72 (0.60, 0.82)	0.74 (0.61, 0.84)	0.02 (-0.12, 0.16)	0.40	1.22 (0.29, 5.36)	1.11 (0.50, 2.57)
Whole egg						
6-month	0.41 (0.30, 0.53)	0.42 (0.29, 0.55)	0.01 (-0.13, 0.15)	0.45	1.08 (0.27, 4.37)	1.04 (0.55, 1.97)
7-month	0.50 (0.38, 0.63)	0.42 (0.29, 0.56)	-0.08 (-0.23, 0.07)	0.85	0.47 (0.10, 2.11)	0.70 (0.34, 1.40)
19-month	0.57 (0.45, 0.69)	0.69 (0.56, 0.80)	0.12 (-0.03, 0.26)	0.06	3.47 (0.74, 18.39)	1.81 (0.87, 4.08)